SingleRAN

IPv6 Transmission Feature Parameter Description

Issue Draft A

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Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
  Changes in functions and their corresponding parameters
- Editorial changes
  Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 02 (2025-06-28).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added support for IPv6 transmission by the UMPThb. For details, see: • **4.3.2 Interface** • **7.6.2 Application on the Base Station**	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for the base station to transmit backup power saving data to microwave devices. For details, see **8.5 LLDP**.	Modified parameter: Added the **BKP_PSAV_EVALUATE_DATA_SW** option to the **LLDPGLOBAL**.***BTSMWCOSW (LTE eNodeB, 5G gNodeB)*** parameter.	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	Base Station Model
Added support for precise reporting of X2/eX2/gNBCUX2/gNBCUXn/gNBDUeXn user-plane fault alarms. For details, see **8.4 GTP-U Echo**.	Modified parameter: Added the **INTER_UP_ALM_PRECISE_SW** option to the **TRANSALGEXTPARA**.***TNLBEARERMNGALGSW (LTE eNodeB, 5G gNodeB)*** parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added X2 interfaces between NR SA and LTE to IPv6 transmission interfaces. For details, see **3.1 IPv6 Transmission Interfaces**.	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

None

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to LTE FDD, LTE TDD, NR, and NB-IoT.

For details about the definition of base stations in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
NR	FOFD-021212	IPv6	**3 IPv6 Transmission Basic Configuration**
LTE TDD	TDLBFD-003003	VLAN Support (IEEE 802.1p/q)	**4.3.3 VLAN**
LTE FDD	LBFD-003003	VLAN Support (IEEE 802.1p/q)	**4.3.3 VLAN**
NB-IoT	MLBFD-12000303	VLAN Support (IEEE 802.1p/q)	**4.3.3 VLAN**
NR	FBFD-010019	VLAN Support (IEEE802.1p/q)	**4.3.3 VLAN**
LTE FDD	LOFD-003017	S1 and X2 over IPv6	**4.6 IPv6 Transport Protocol Stacks for Interfaces**
LTE TDD	TDLOFD-003017	S1 and X2 over IPv6	**4.6 IPv6 Transport Protocol Stacks for Interfaces**
NB-IoT	MLOFD-003017	S1 over IPv6	**4.6 IPv6 Transport Protocol Stacks for Interfaces**
LTE FDD	LBFD-003006	IPv4/IPv6 Dual Stack	**6 IPv4/IPv6 Dual-Stack Networking for Interfaces**
LTE TDD	TDLBFD-003006	IPv4/IPv6 Dual Stack	**6 IPv4/IPv6 Dual-Stack Networking for Interfaces**
NB-IoT	MLBFD-12000307	IPv4/IPv6 Dual Stack	**6 IPv4/IPv6 Dual-Stack Networking for Interfaces**
NR	FBFD-021101	IPv4/IPv6 Dual Stack	**6 IPv4/IPv6 Dual-Stack Networking for Interfaces**
LTE FDD	LBFD-00202103	SCTP Multi-homing	**7.2 Control-Plane SCTP Multihoming**
LTE TDD	TDLBFD-00202103	SCTP Multi-homing	**7.2 Control-Plane SCTP Multihoming**

RAT	Feature ID	Feature Name	Chapter/Section
NB-IoT	MLBFD-12000419	SCTP Multi-homing	**7.2 Control-Plane SCTP Multihoming**

3 IPv6 Transmission Basic Configuration

3.1 IPv6 Transmission Interfaces

Interfaces that support IPv6 transmission include:

- S1, X2, and eX2 interfaces of LTE
- S1, X2, and eXn interfaces of non-standalone (NSA) NR
- NG, Xn, and eXn interfaces of standalone (SA) NR
- OM channel interface and clock synchronization interface of a base station

For the position of each interface on a network, see *IPv4 Transmission*.

3.2 Transmission Configuration Model

IPv6 transmission or IPv4/IPv6 dual-stack transmission supports only the new transmission configuration model.

When **GTRANSPARA**.***TRANSCFGMODE (5G gNodeB, LTE eNodeB)*** is set to **NEW**, the new transmission configuration model is used. The transmission model is independent from the device model. Only the **INTERFACE** MO needs to be configured. The cabinet, subrack, or slot number does not need to be configured. This facilitates the extension of new transmission functions and reduces transmission configuration parameters.

NOTE

For details about how to configure transmission resource management and IPsec, see *Transmission Resource Management* and *IPsec*.

The transmission configuration models of different RATs in a co-MPT multimode base station must be the same. When the co-MPT multimode base station is configured with IPv4/IPv6 dual-stack transmission, the old IPv4 transmission model on the live network needs to be reconstructed. For example, if NR is superimposed on LTE, then NR uses IPv6 transmission and LTE still uses IPv4 transmission. In this case, the old IPv4 transmission model of LTE needs to be reconstructed into the new transmission model. The northbound tool must support the new IPv4 and IPv6 transmission models. The old IPv4 transmission model of the northbound tool must be reconstructed into the new IPv4 transmission model.

A separate-MPT multimode base station can be considered as multiple single-mode base stations. The transmission configuration model of each RAT is independent. The RAT that uses IPv4 transmission can use either the old or new model.

4 Transmission Protocol Stacks

4.1 Overview

This chapter describes the basic protocols for IPv6 transmission at the physical, data link, network, and transmission layers, as well as the protocol stacks of IP transmission interfaces.

The physical layer protocols for IPv6 and IPv4 transmission are the same on the base station side. The data link layer, network layer, and transmission layer protocols for IPv6 transmission are different from those for IPv4 transmission. This chapter describes the differences between IPv6 and IPv4 transmission protocols. For details about other information, see *IPv4 Transmission*.

4.2 Physical Layer

The physical layer is the bottom layer of the general transmission protocol stack and is identical between IPv4 and IPv6 transmission. The main physical layer port involved in base station IPv6 transmission is the Ethernet port.

4.3 Data Link Layer

The data link layer is the second layer of the Transmission Control Protocol/ Internet Protocol (TCP/IP) protocol stack. It receives and processes data from the physical layer and provides reliable data transmission for the network layer. Base station IPv6 transmission supports only Ethernet ports.

4.3.1 MAC

IPv6 and IPv4 use the same MAC address and frame format. Ethernet has multiple MAC frame formats. The Ethernet II frame format is defined in RFC 894, as shown in **Figure 4-1**. The upper-layer protocol type value 0x86DD in the Ethernet frame indicates the IPv6 protocol.

Figure 4-1 Ethernet II frame format

Frame header

Destination MAC address	Source MAC address	Type	Data	FCS
6 bytes	6 bytes	2 bytes	46-1500 bytes	4 bytes

4.3.2 Interface

An interface can be configured for a physical port (for example, an Ethernet port or an Ethernet trunk group) or for a logical port (loopback interface) on a base station. Interfaces are classified into common interfaces and VLAN sub-interfaces based on whether VLAN tags are contained in packets. The **INTERFACE**.***ITFTYPE (5G gNodeB, LTE eNodeB)*** parameter specifies the interface type.

- Common interfaces can be configured for a physical port or a logical interface on a base station. Packets sent from a common interface do not contain VLAN tags. If the link type of the port on the peer transmission device connected to the base station is set to **Access** or **Hybrid**, packets sent from the peer transmission device to the base station do not contain VLAN tags.

  NOTE

  IPv6 does not support VLAN configuration using the **VLANMAP** and **VLANCLASS** MOs.

- VLAN sub-interfaces must be configured for a physical port. Packets sent from or received by a VLAN sub-interface contain VLAN tags. Each VLAN sub-interface is planned as a VLAN and corresponds to only one VLAN ID. The base station attaches VLAN tags to traffic flows before sending the traffic flows to a VLAN interface.

  Interfaces are configured in the **INTERFACE** MO on the base station and shared by IPv4 and IPv6 transmission. IPv4 transmission is supported by default. To enable IPv6 transmission, set the **INTERFACE**.***IPV6SW (5G gNodeB, LTE eNodeB)*** parameter to **ENABLE**. IPv6 transmission must be first enabled for an interface and then IPv6 addresses or IPv4/IPv6 dual-stack addresses can be added to the **INTERFACE** MO.

NOTE

According to the hardware type, the maximum size of a data packet that can be received over an Ethernet port is as follows:

- BBU:

  BBU5900C: 9320 bytes

- Main control boards:

  UMDU: 2000 bytes

  UMPTb: 2000 bytes

  UMPTe: 2024 bytes

  UMPTg/UMPTga: 9320 bytes

  UMPTha/UMPThb: 9320 bytes

- Transmission extension board:

  UTRPc: 2000 bytes

4.3.3 VLAN

IPv6 transmission has the same VLAN format and functions as IPv4 transmission. However, VLAN configuration modes supported on the base station side are different between IPv6 and IPv4 transmission. **Table 4-1** lists the VLAN configuration requirements for base stations using IPv4, IPv6, or IPv4/IPv6 dual-stack transmission.

Table 4-1 VLAN configuration requirements for different transmission types

Transmission Type	Transmission Configuration Model	VLAN Requirement
IPv6 transmission	New model	The VLAN works in interface VLAN mode.
IPv4 transmission	New model	The VLAN works in single VLAN mode, group VLAN mode, or interface VLAN mode.
	Old model	The VLAN works in single VLAN mode or group VLAN mode.
IPv4/IPv6 dual-stack transmission	New model	VLANs for both IPv4 and IPv6 transmission work in interface VLAN mode. IPv6 and IPv4 transmission can use the same or different VLAN IDs.

In IPv6 or IPv4/IPv6 dual-stack transmission, VLANs can be configured only in interface mode. A VLAN includes VLAN ID and VLAN priority.

- A VLAN ID is specified by the **INTERFACE**.***VLANID (5G gNodeB, LTE eNodeB)*** parameter.
- The DSCP value and VLAN priority are specified by the **DSCP2PCPREF.*DSCP*** and **DSCP2PCPREF.*PCP*** parameters, respectively. The **INTERFACE**.***DSCP2PCPMAPID*** parameter specifies a mapping relationship between DSCP values and VLAN priorities in the DSCP-PCP mapping list (**DSCP2PCPMAP** MO).

4.4 Network Layer

The network layer uses IP as the principal protocol and is also referred to as the IP layer. IPv6 is the second-generation standard network layer protocol and has the following improvements based on the first generation:

- More IP addresses

  IPv6 increases the IP address size from 32 bits to 128 bits, and therefore provides many more IP addresses.
- Auto-configuration mechanism

IPv6 supports auto-configuration of addresses. That is, a host automatically detects network provisioning and obtains the IPv6 address assigned to the host.

- Enhanced transmission security

  IPv6 standard extension headers are used for IPsec. IPv6 nodes support IPsec, which provides end-to-end security.

During the IPv4-to-IPv6 transition, the two protocols will coexist. Base stations support IPv4/IPv6 dual-stack transmission to ensure a smooth evolution from IPv4 to IPv6.

During IPv4/IPv6 dual-stack transmission, a dual-stack node communicates with an IPv4 node using the IPv4 protocol, whereas it communicates with an IPv6 node using the IPv6 protocol.

4.4.1 Basic IPv6 Information

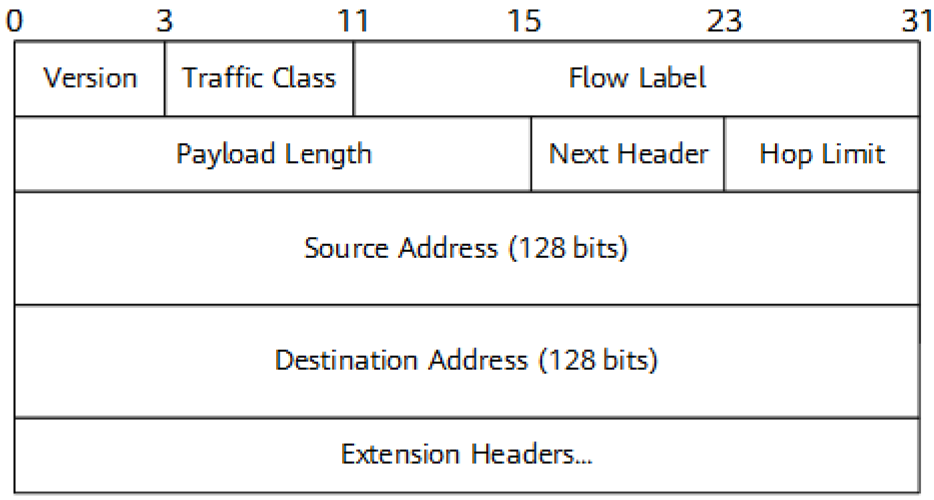
4.4.1.1 IPv6 Packet Format

An IPv6 packet consists of three parts: basic packet header, extension packet header, and upper-layer protocol data unit.

Basic Packet Header

A basic IPv6 packet header occupies 40 bytes, including basic information about packet forwarding. This header is mandatory for IPv6 packets. **Figure 4-2** shows the format of a basic IPv6 packet header.

Figure 4-2 Format of the basic IPv6 packet header



The fields in an IPv6 packet header are described as follows:

- Version: This field indicates the IP protocol version and occupies four bits. The value is **6** for IPv6 transmission.
- Traffic Class: This field occupies eight bits and identifies the class or priority of IPv6 packets. This field is similar to the Type of Service (TOS) field for IPv4 transmission.

- Flow Label: This field occupies 20 bits and indicates the specific packet sequence which a packet belongs to between the source and the destination nodes. This field needs to be processed by the intermediate IPv6 router. Generally, a flow can be determined by the IPv6 address and flow label of the source node or destination node.

  For GTP-U packets, the Flow Label field is filled with the last 20 bits of the tunnel endpoint identifier (TEID).

  For UDP packets on the user plane, the Flow Label field is filled with the last 20 bits of the hash value of the source port number and destination port number.

  For other IP packets, the Flow Label field is filled with 0.
- Payload Length: This field occupies 16 bits and specifies the length of the valid payload of an IPv6 packet. A valid payload is the part that follows the basic IPv6 packet header in a packet. A valid payload includes the extension packet header and upper-layer protocol data unit.
- Next Header: This field occupies eight bits and specifies the type of the extension header that follows the IPv6 packet header or the type of the transmission layer protocol. If a packet has an extension header, this field identifies the header following the IPv6 header. If the packet has no extension header, this field specifies the transport layer protocol type. For example, the value **17** indicates UDP and the value **6** indicates TCP.
- Hop Limit: This field occupies eight bits and specifies the maximum number of hops allowed by IPv6 packets. The value of this field decreases by 1 each time the packet is forwarded by a router. A router will discard a packet if it finds out that the value of this field in the packet is 0.
- Source Address: This field occupies 128 bits and specifies the source IP address of packets.
- Destination Address: This field occupies 128 bits and specifies the destination IP address of packets.

Extension Packet Header

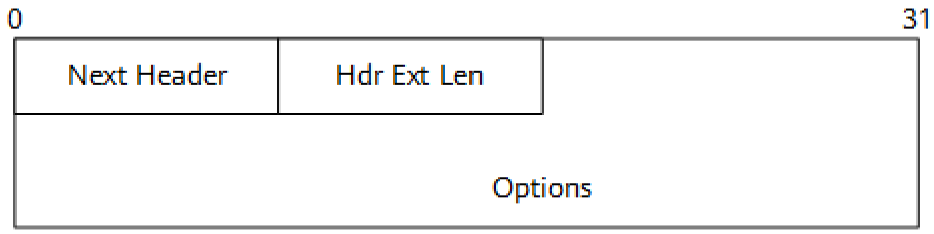
IPv6 reduces the payload of the header by moving the optional fields to the extension packet header. This improves the packet handling efficiency of the network device. The extension packet header is optional and is used for specific functions. After receiving an IPv6 packet whose destination IP address is the IP address of the local base station, the base station checks the IPv6 extension header. A packet sent by the peer device (router, core network, base station, clock server, or security gateway) must contain only the following extension headers. Otherwise, the packet is discarded by the base station. When forwarding IPv6 packets whose destination IP address is not the IP address of the local base station, the base station can only forward the packets based on routes and does not check or process IPv6 extension headers.

- Hop-by-hop options header

  This header is used for multicast snooping discovery protocol packets. Base stations can contain this header only in the multicast listener discovery (MLD) protocol packets.

  This header specifies the sending parameters for each hop on the transmission path. Each intermediate node on the transmission path needs to read and process this header. The value of the Next Header field is **0**. **Figure 4-3** shows the format of the hop-by-hop options header.

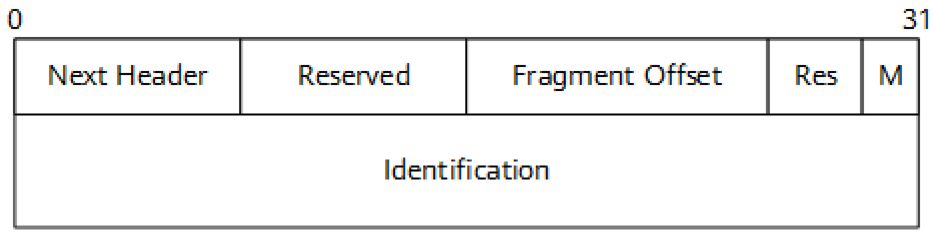
Figure 4-3 Format of the hop-by-hop options header



- Fragment header

  A packet needs to be fragmented if it exceeds the maximum transmission unit (MTU). A fragment header is used for packet fragmentation. IPv6 intermediate nodes do not allow for IPv6 packet fragmentation. IPv6 packets must be fragmented by the source node. The value of the Next Header field of the fragment header is **44**. **Figure 4-4** shows the format of the fragment header.

Figure 4-4 Format of the fragment header.



The fields in this header are described as follows:

– Next Header: This field occupies eight bits and specifies the value of the next packet header. This value is the type value of the next extension header or transmission layer protocol.

– Reserved: The field occupies eight bits. The initial value is **0**. This field is ignored during reception.

– Fragment Offset: This field occupies 13 bits and specifies the offset of the current packet content in the original packet.

– Res: This field occupies two bits. The initial value is **0**. This field is ignored during reception.

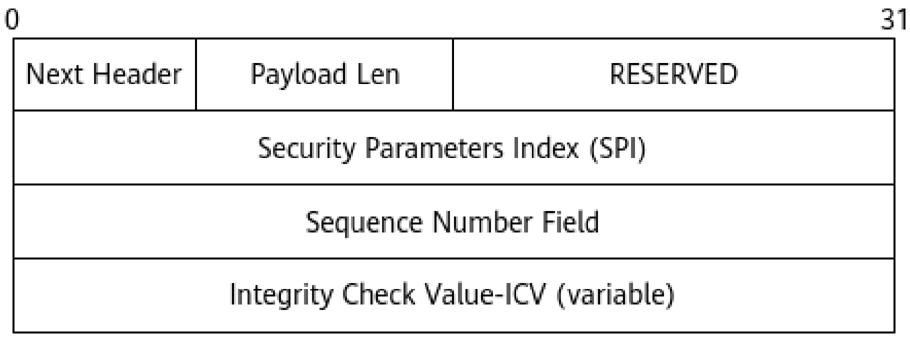
– M: This field occupies one bit. The value **1** indicates that there are subsequent fragments. The value **0** indicates that the current packet is the last fragment.

– Identification: This field occupies 32 bits. The source node allocates an ID to each packet fragment. This ID identifies the packet to which a fragment belongs. The destination node assembles packet fragments based on this ID. The ID of a packet fragment must be unique. The value of the Identification field is incremented by one for one more packet fragment. A 32-bit value ensures that each packet fragment can have a unique ID.

A fragment with the Fragment Offset and the M flag both set to 0 is referred to as an atomic fragment. The base station does not support the processing of packets containing atomic fragments and will discard such packets.

- Authentication header: This header is used to ensure IP security and provides authentication services. The value of the Next Header field is **51**. This header is the same in IPv4 and IPv6. **Figure 4-5** shows the header format.

Figure 4-5 Format of the authentication header



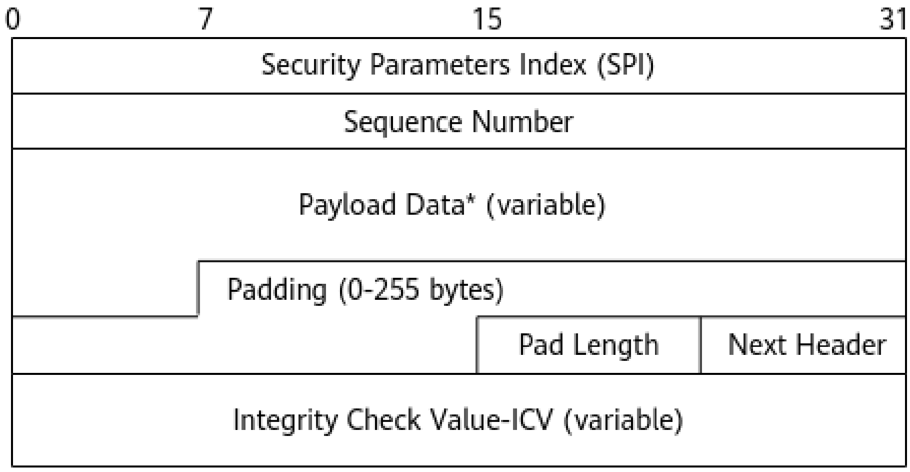
The fields in this header are described as follows:

  - Next Header: This field occupies eight bits and specifies the value of the next packet header. This value is the type value of the next extension header or transmission layer protocol.
  - Payload Len: This field occupies eight bits and specifies the length of the authentication extension header.
  - RESERVED: This field occupies 16 bits. The initial value is **0**. This field is ignored during reception.
  - Security Parameters Index: This 32-bit field can be set to any value. This field, the destination IP address, and security protocol AH uniquely identify the security status of packets. The value **0** is reserved locally. The values **1** to **255** are reserved for Internet Assigned Numbers Authority (IANA).
  - Sequence Number: This field occupies 32 bits. The value of this field is incremented by one each time an AH packet is sent. The initial value is **0**.
  - Integrity Check Value-ICV: It is variable in length and is used to check the integrity of a packet. The value must be an integral multiple of 32 bits. If the length is insufficient, padding is required.

- Encapsulating security payload (ESP)

  An ESP ensures IP security. The value of the Next Header field is **50**. Similar to the authentication header, an ESP is the same for both IPv4 and IPv6 transmission. An ESP is also referred to as an IPsec ESP. For descriptions of the fields in an ESP, see the descriptions of fields in the authentication header.

Figure 4-6 Format of the ESP



Upper-Layer Protocol Data Unit

An upper-layer protocol data unit includes the upper-layer protocol packet header and its valid payload, such as the TCP and UDP. This part is the same as that in an IPv4 packet.

4.4.1.2 IPv6 Address

4.4.1.2.1 Format of IPv6 Addresses

An IPv6 address is a 128-bit value in the format of X:X:X:X:X:X:X:X.

- A 128-bit IPv6 address is divided into eight groups. The 16 bits in each group are represented by four hexadecimal characters (0 to 9 and A to F). The groups are separated by colons (:). Each X represents a group of hexadecimal numbers. For example, 2001:db8:130f:0000:0000:09c0:876a:130b is an IPv6 address. For the convenience of writing, the leading zeros in each group can be omitted. This address can be written as 2001:db8:130f:0:0:9c0:876a:130b.

  Two or more consecutive groups in which all values are 0 in an IPv6 address can be replaced with double colons (::). In this way, the length of an IPv6 address can be reduced. The preceding address can be written as 2001:db8:130f::9c0:876a:130b.
- In an IPv6 address, only one double colon (::) can be used. Otherwise, the number of 0 in each segment cannot be determined when the compressed address is restored to a 128-bit value.

4.4.1.2.2 IPv6 Address Types

There are three types of IPv6 addresses: unicast addresses, multicast addresses, and anycast addresses. Compared with IPv4, IPv6 does not support broadcast addresses. Instead, IPv6 uses multicast addresses. In addition, IPv6 supports anycast addresses.

- A unicast address identifies a network interface of a transmission node. If the destination address of an IP packet is a unicast address, this packet is sent to the network interface identified by this unicast address.

- A multicast address identifies a group of interfaces that belong to different nodes. An IPv6 multicast address is similar to an IPv4 broadcast address. Packets destined for a multicast address are sent to all interfaces identified by this multicast address. Compared with IPv4, IPv6 does not support broadcast addresses. Broadcast addresses are replaced by multicast addresses.
- An anycast address identifies a group of interfaces, which generally belong to different nodes. Packets sent to an anycast address are transmitted to an interface nearest to the source node in a group of interfaces identified by this address. The "nearest" interface is measured based on the distance specified by the routing protocol.

NOTE

Base stations support only unicast and multicast addresses, and do not support anycast addresses.

Table 4-2 lists the IPv6 address types supported by a base station.

Table 4-2 IPv6 address types and configurations supported by a base station

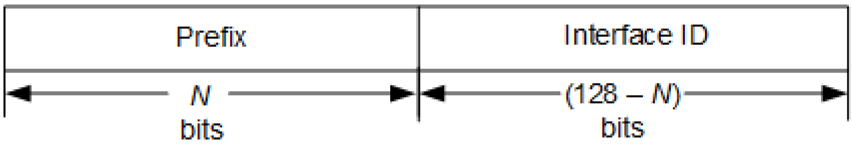
Category	Sub-category	Description	Parameter Setting on the Base Station
Unicast address	Globally scoped unicast address	The global unicast address and unique local unicast address are included. This address is a service IP address or interface IP address. It is configured for an Ethernet port, an Ethernet trunk, or a loopback interface.	It is configured using the **ADD IPADDR6** command.
	Link-local address	This address is used in many IPv6 protocols, such as the Neighbor Discovery Protocol (NDP), DHCPv6 protocol, and multicast listener discover (MLD) protocol.	It is automatically generated by the base station and cannot be configured.
		This is an IP address for local maintenance.	It is configured using the **SET LOCALIP6** command.
Multicast address	All-node multicast address	This address is used in NDP.	The base station automatically adds all-node multicast addresses.

Category	Sub-category	Description	Parameter Setting on the Base Station
	Solicited-node multicast address	This address is used in NDP.	After a unicast address is configured or automatically generated on an Ethernet interface, the base station automatically generates a solicited-node multicast address.

Unicast Address

A unicast address consists of a subnet prefix and an interface ID, as shown in **Figure 4-7**. A subnet prefix is associated with a link. An interface ID identifies a link interface. A unicast address must be unique in a subnet.

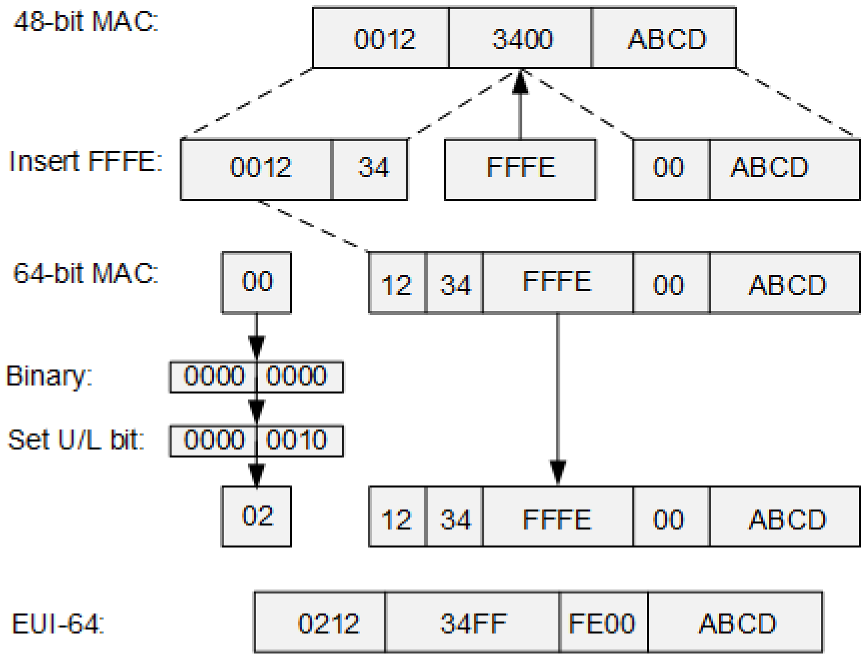
Figure 4-7 Unicast address format



- Network prefix: This field occupies *N* bits. It is equivalent to the network ID in an IPv4 address.
- Interface ID: This field occupies 128–*N* bits. It is equivalent to the host ID in an IPv4 address.

A unicast address can be derived from a link-layer MAC address or be manually planned. An IPv6 address in the IEEE EUI-64 format is converted from a MAC address at the link layer. The interface ID occupies 64 bits. To convert a 48-bit MAC address into a 64-bit interface ID, insert the hexadecimal number FFFE (1111 1111 1111 1110) into the MAC address. Then, set the U/L bit (the seventh bit starting from the most significant bit) to 1 to obtain the interface ID in EUI-64 format. **Figure 4-8** shows the conversion process.

Figure 4-8 Converting a MAC address to an interface ID in EUI-64 format

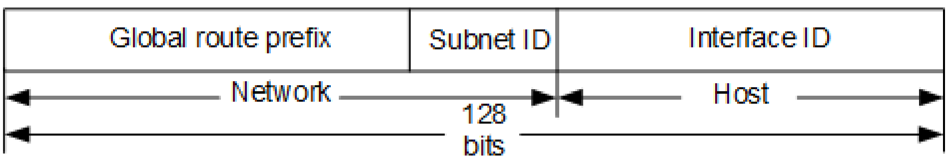


IPv6 defines several types of unicast addresses. Base stations support the following types of unicast addresses:

- Global unicast address

  A global unicast address is an address that can be used for IP routing globally, like a public network address in IPv4. The subnet prefix of a global unicast address must be unique. **Figure 4-9** shows the global unicast address format. The base station can only use a global unicast address as the service IP address to communicate with the S-GW, MME, OSS, and IP clock server. IPv6 addresses are manually configured in **IPADDR6** MOs. The subnet prefix length in an IPv6 address is specified by the **IPADDR6**.***PFXLEN (5G gNodeB, LTE eNodeB)*** parameter. An IPv6 address is specified by the **IPADDR6**.***IPV6 (5G gNodeB, LTE eNodeB)*** parameter.

  Figure 4-9 Global unicast address format

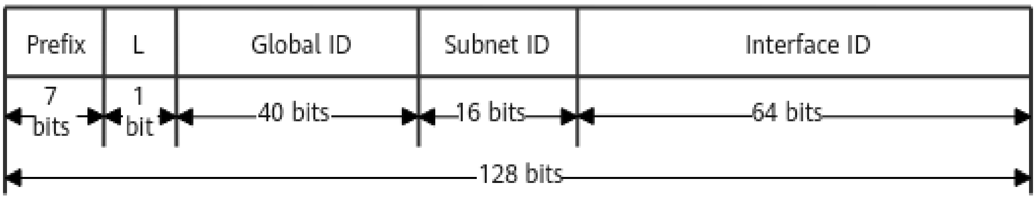
  Global unicast addresses are organized into a three-level hierarchy:

  – Global route prefix: assigned by the provider to an organization (end site). The first three digits of a global route prefix allocated by the Internet Assigned Numbers Authority (IANA) are 001.
  – Subnet ID: used for subnet division.
  – Interface ID: uniquely identifies an interface.

- Unique local unicast address

A unique local unicast address is a type of IPv6 address. It contains a global ID in the address structure. In the RFC standard, the unique local unicast address and the global unicast address are collectively called globally scoped unicast address. A unique local unicast address cannot be routed globally and can only be used for communication between sites on the local network. It is equivalent to a private IPv4 address. Routers on the local network must be able to forward IPv6 packets whose source or destination address is a unique local unicast address. The peer device must also support a unique local unicast address. The unique local unicast address on the base station is configured in an **IPADDR6** MO, which is the same as the global unicast address.

Figure 4-10 Unique local unicast address format



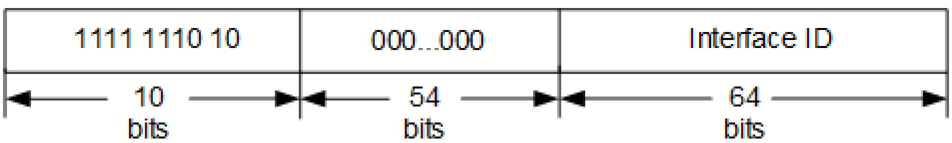
As shown in **Figure 4-10**:

- Prefix: fc00:: /7
- L: The value 1 indicates local assignment, and 0 is reserved for future use.
- Global ID: used to create a globally unique prefix. RFC 4193 provides a method for creating a global ID.
- Subnet ID: used to identify a subnet of the local site.
- Interface ID: uniquely identifies an interface.

- Link-local address

  The prefix of a link-local address is fe80:: /10. A link-local address is used for communication between local link nodes in compliance with NDP, DHCPv6, and MLD. IPv6 packets whose destination address is a link-local address are not forwarded by a router. Each interface on a network node must have a link-local address. **Figure 4-11** shows the link-local address format.

Figure 4-11 Link-local address format



The base station supports automatic generation of link-local addresses for Ethernet ports or Ethernet trunks. The interface ID of the link-local address of a physical interface is generated in IEEE EUI-64 format based on the MAC address and cannot be manually configured in an **IPADDR6** MO. A link-local address generated based on a MAC address is globally unique. A link-local address can be set as the IPv6 address of the local maintenance channel using the **LOCALIP6**.***IP6 (5G gNodeB, LTE eNodeB)*** parameter locally.

- Unassigned address

  An unassigned address is also called an all-zero address, which is 0:0:0:0:0:0:0:0/128. This address is not assigned to any interface and cannot be used as the destination address of packets. The unassigned address can be used as the source address of packets only when a node is started and no IP address is assigned to an interface. For example, the unassigned address can be used as the source address of Neighbor Solicitation (NS) packets during address conflict detection.

- Loopback address

  The loopback address is 0:0:0:0:0:0:0:1/128. It is used within a node and is not allocated to any interface. It is equivalent to the loopback address 127.0.0.1 in IPv4. If the destination address of an IP packet is a loopback address, this packet is sent to the sender itself.

- IPv4-mapped IPv6 address

  The address format is 0:0:0:0:0:ffff:*X.X.X.X*. Here, *X.X.X.X* is a 32-bit IPv4 address. It is used for applications that support only IPv6 to work on dual-stack nodes.

NOTE

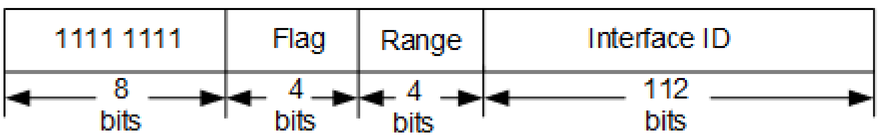
- In IPv6 networking, base stations support only global unicast addresses and unique local unicast addresses.
- In this version, IPv4-mapped IPv6 addresses are not supported.
- The prefix length of the IP address of an Ethernet interface can be set to 127 bits. Note that during transmission network planning, all 0s and all 1s cannot be used in other network segments except the directly connected 127-bit network segment.
- In RFC 3513, IPv6 addresses with the prefix of 0xFEC0/10 are defined as site-local addresses. However, this address type is discarded in RFC 3879. Some devices may still use IPv6 addresses with the prefix of 0xFEC0/10 as site-local addresses. To avoid confusion on the live network and impact on services, IPv6 addresses with the prefix of 0xFEC0/10 are not supported.

Multicast Address

A multicast address identifies a group of network interfaces, which usually belong to different nodes. If the destination address of a packet is a multicast address, the packet is sent to all the network interfaces in the multicast group. Multicast addresses in IPv6 transmission serve the role of broadcast addresses in IPv4 transmission. Multicast addresses are used for one-to-many communication. The eNodeB/gNodeB does not support the configuration of multicast addresses, but can receive and parse packets with a multicast address as the destination address.

Figure 4-12 shows the multicast address format.

Figure 4-12 Multicast address format



Wherein:

- The most significant eight bits are always 1.
- Flag: 4 bits, indicating whether the multicast address is a fixed multicast address. The value **0000** indicates that this address is permanent. The value **0001** indicates that this address is temporary.
- Range: 4 bits, indicating the range of the multicast address.
  - 1: interface-local range
  - 2: link-local range
  - 5: site-local range
  - 8: organization-local range
  - E: global range
- Group ID: identifies a multicast group.

ff00::/16 to ff0f::/16 are reserved multicast addresses and cannot be allocated by multicast groups. They serve the role of the broadcast function of IPv4 transmission.

- An all-node multicast address (ff02::1) is the destination address of NDP Router Advertisement (RA) packets. All nodes join this multicast group to receive packets whose destination address is this multicast address.

  The trigger conditions for generating the all-node multicast address are as follows:
  - An **INTERFACE** MO is added and the **INTERFACE**.***IPV6SW (5G gNodeB, LTE eNodeB)*** parameter is set to **ENABLE**.
  - An **INTERFACE** MO with the **INTERFACE**.***IPV6SW (5G gNodeB, LTE eNodeB)*** parameter set to **ENABLE** has been configured.

  The trigger conditions for deletion are as follows:
  - The **INTERFACE** MOs with the **INTERFACE.*IPV6SW (5G gNodeB, LTE eNodeB)*** parameter set to **ENABLE** are deleted.
  - The **INTERFACE**.***IPV6SW (5G gNodeB, LTE eNodeB)*** parameter is set to **DISABLE**.
- An all-router multicast address (ff02::2) is the destination address of NDP Router Solicitation (RS) packets. All routers join this multicast group to receive packets whose destination address is this multicast address.
- A solicited-node multicast address is ff02::1:ff*XX*:*XXXX*. *X* indicates the least significant 24 bits of the generated unicast address or anycast address. For example, if the unicast address is 2001:db8::800:200e:8c6c, the automatically generated solicited-node multicast address is ff02::1:ff0e:8c6c. After a unicast or anycast address is configured or automatically generated for an interface of a node, a solicited-node multicast address is automatically generated. On a base station, multiple unicast addresses can be mapped to one solicited-node multicast address. When a unicast address is configured on an interface, a solicited-node multicast address is automatically generated if there is no corresponding solicited-node multicast address. When a unicast address is deleted, the corresponding solicited-node multicast address is deleted only when the unicast address deleted is the last one corresponding to the solicited-node multicast address. This address is used as the destination address of NDP address resolution or address conflict detection packets.

Anycast Address

An anycast address identifies a group of network interfaces, which belong to different nodes. If a physical interface of a node is configured with an anycast address, the node is called an anycast member. The information about anycast members must be sent to relevant routers on the network. Irrelevant routers do not need to learn this information.

NOTE

Base stations do not support anycast addresses.

If the destination address of an IP packet is an anycast address, the packet is forwarded to the "nearest" anycast member rather than to each anycast member. This is the most significant difference between the anycast service and multicast service.

The anycast address format is the same as the unicast address format, as shown in **Figure 4-9**. RFC 4291 defines a subnet router anycast address with the interface ID of 0.

4.4.1.3 IPv6 Static Routes

The principle for forwarding IPv6 static routes is the same as that for forwarding IPv4 static routes. Base stations support static destination-based routing and source-based routing in IPv6 transmission.

- The packet sending principle of static destination-based routing in IPv6 transmission is the same as that in IPv4 transmission. Base stations search for the egress port and gateway IP address based on the destination IP address of a packet. The following routes are supported:
  - Active/standby route: Static routes with the same IPv6 destination address and prefix length but different priorities function as the active and standby routes. Packets are forwarded using the route with the highest priority.
  - Direct route: This is a type of destination route. A base station automatically generates the direct route whose destination address is located in the network segment of the interface IP address. A direct route does not require manual maintenance and has the highest priority.
  - Host route: This is a manually configured route with a 128-bit prefix.
  - Network segment route: The manually configured prefix length is not 128 bits.
  - Default route: The destination route of a default route is an all-zero value.

  NOTE

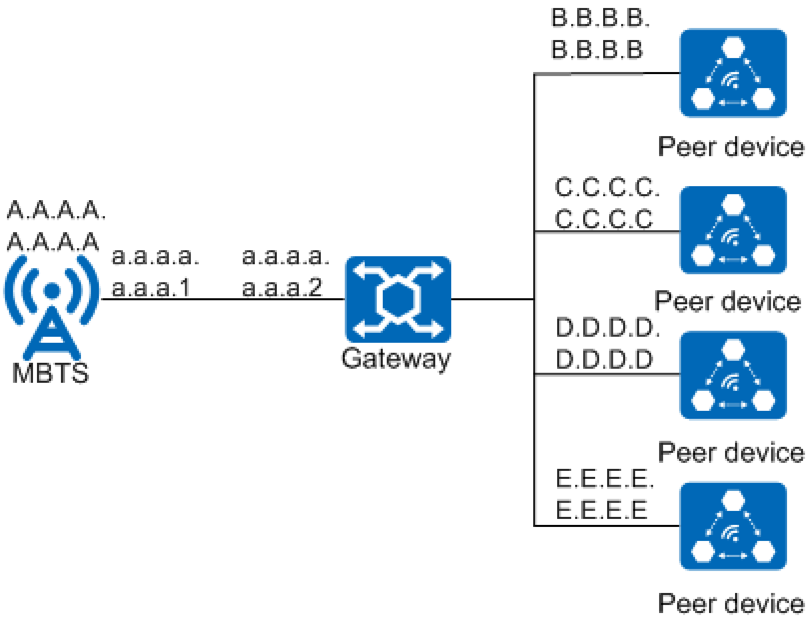
  Base station IPv6 transmission does not support equal-cost routes with the same priority.

  Base station IPv6 static destination-based routings are configured in the new transmission model. IPv6 static route entries are configured in the **IPROUTE6** MO. The **DSP IPROUTE6** command output displays routes taking effect on a base station.
- Static source-based IP routing: With this mechanism, IP addresses of egress ports and gateways for IPv6 packets are determined based on the source IP

addresses. The base station supports source-based routing and source-based routing backup. If the source IP addresses of two routes are the same but the next-hop IP addresses of the two routes are different, the two routes work in active/standby mode when they have different priorities. Packets are forwarded using the route with the highest priority. Base station IPv6 source-based routing does not support equal-cost routes with the same priority. Base station IPv6 static source-based routings can be configured only in the new transmission model. IPv6 static source-based routing entries are configured in the **SRCIPROUTE6** MO. The **DSP SRCIPROUTE6** command output displays source-based routings taking effect on a base station.

Figure 4-13 shows the recommended scenario for source-based routing supported by the base station. Routes to be configured on a base station have the same service IP address but different next-hop IP addresses and different destination IP addresses. In this case, source-based routing is recommended for simplifying route configurations on the base station side.

Figure 4-13 Recommended scenario for source-based routing on the base station



For example:

– In the MME pool scenario, the S1 interface of the base station is connected to multiple peer IP addresses.

– In the core network capacity expansion and migration scenarios, the peer IP address of the S1 interface on the base station changes dynamically.

– In the X2/Xn interface networking and base station L3 networking, multiple peer IP addresses are interconnected to multiple base stations. It is recommended that the X2/Xn interface should use one IP address and the X2/Xn and S1 interfaces should use the same next hop in L3 networking.

If there are multiple service IP addresses on the base station side, the next-hop IP addresses are different, and there is only one destination IP address on the peer end, source-based routing must be used on the base station. If there is only one service IP address on the base station side, the next-hop IP addresses are different, and there are multiple destination IP addresses on the peer end, source-based routing is not supported for the base station.

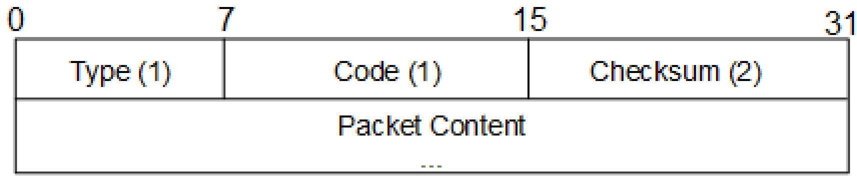
NOTE

If both the destination-based and source-based routings are configured, only the source-based routing takes effect.

4.4.2 ICMPv6

Internet Control Message Protocol for the Internet Protocol Version 6 (ICMPv6) is one of the basic IPv6 protocols. ICMPv6 packets include error packets and information packets. IPv6 neighboring nodes use these packets to report errors and information during packet processing. For details, see RFC 4443. **Figure 4-14** shows the format of an ICMPv6 packet.

Figure 4-14 ICMPv6 packet format



The fields in the packet are described as follows:

- Type: This field specifies the packet type. The values from **0** to **127** indicate error packets. The values from **128** to **255** indicate information packets.
- Code: This field specifies the subtype of packets.
- Checksum: This field specifies the checksum of ICMPv6 packets.

4.4.2.1 Types of ICMPv6 Error Packets

ICMPv6 error packets are classified into the following types:

- Destination Unreachable Message

  If an IPv6 node finds that no route is available to the destination or that the destination address or port is unreachable when forwarding an IPv6 packet, it sends an ICMPv6 Destination Unreachable Message to the source node.
- Packet Too Big Message

  If an IPv6 node finds that the packet size exceeds the path MTU of the egress port when forwarding an IPv6 packet, it sends an ICMPv6 Packet Too Big Message to the source node. The error packet contains the path MTU of the egress port.
- Time Exceeded Message

  During IPv6 packet sending and receiving, when the Hop Limit value decreases to 0, the device sends an ICMPv6 Time Exceeded Message to the source node sending the IPv6 packet. If packet fragmentation or assembling times out, an ICMPv6 Time Exceeded Message is generated.
- Parameter Problem Message

A destination node checks the validity of a received IPv6 packet. If a field in the IPv6 basic header or extension header is incorrect, the NextHeader value is unknown, or the extension header has unknown options, an ICMPv6 Parameter Problem Message is sent to the source node.

4.4.2.2 Types of ICMPv6 Information Packets

ICMPv6 information packets are classified into request information (Echo Request) and response information (Echo Reply). ICMPv6 packets are used for network fault diagnosis, PMTU discovery, and neighbor discovery. During the connectivity test between two nodes, the node that receives the Echo Request packet returns an Echo Reply packet to the source node. In this way, packets are transmitted and received between the two nodes.

4.4.3 Neighbor Discovery

Neighbor discovery is a group of messages and processes that determine the relationship between neighboring nodes. Neighbor discovery implements the functions in **4.4.3.2 Address Resolution**, **4.4.3.4 Duplicate Address Detection**, **4.4.3.6 Router Discovery**, and **4.4.3.7 Route Redirection** using ICMPv6 information packets.

4.4.3.1 Neighbor Discovery Packet Format

Neighbor discovery defines five types of ICMPv6 messages. **Table 4-3** lists the types and functions of these messages.

Table 4-3 Types and functions of ICMPv6 messages

Message Type	Function
Neighbor solicitation (NS)	• Obtains the data-link-layer address of a neighbor. • Detects address conflicts. • Checks whether the neighbor is reachable.
Neighbor advertisement (NA)	• Responds to an NS message. • If the data-link-layer address of a node changes, the node sends an unsolicited NA message to inform the neighbor of the change.
Router solicitation (RS)	Requests the prefix and other configuration information for automatic configuration of the host node after the host node is started.

Message Type	Function
Router advertisement (RA)	● Responds to an RS message. ● If RA messages are not suppressed, the router periodically sends RA messages, which include prefix information option and flag bits. **NOTE** Whether a router sends information contained in RA messages needs to be configured on the router. Routers must be deployed on the network.
Redirection packet	If certain conditions are met, the default router sends a redirection packet to the source host so that the host selects the correct next hop address to send subsequent packets. **NOTE** Routers must be deployed on the network. The base station does not support the redirection function and discards received redirection messages.

- Neighbor solicitation (NS)

  An IPv6 node obtains the data-link-layer address of the neighbor from an NS packet, checks whether the neighbor is reachable, and performs duplicate address detection. **Figure 4-15** shows the NS packet format.

  Figure 4-15 NS packet format

  – Type: The value is **135**.
  – Code: The value is **0**.
  – Checksum: This is the checksum field.
  – Reserved: The sender must initialize the field value to **0**. The receiver must ignore this field.
  – Target Address: This field specifies the IP address of the solicited destination. It cannot be a multicast address. It can be a local-link, local-site, or global address.
  – Options: This field specifies the data-link-layer address of the packet sender. Only address options at the source data link layer can be used in NS packets. If the source address of an IPv6 header is an unassigned address, this field cannot be contained.
- NA: An NA packet is a response of an IPv6 node to an NS packet. An IPv6 node can proactively send an NA packet when the data link layer changes to

notify the adjacent node that its data-link-layer address or role changes. **Figure 4-16** shows the NA packet format.

Figure 4-16 NA packet format

- Type: The value is **136**.
- Code: The value is **0**.
- Checksum: This is the checksum field.
- R: This field is the router flag and indicates the role of the NA packet sender. The value **1** indicates that the sender is a router. The value **0** indicates that the sender is a host. The router flag is used for NUD to determine when a router becomes a host.
- S: This field is the solicited flag, which is a response to the NA packet sent by the unicast NS. In NUD, S is a neighbor reachability acknowledgment flag. During the multicast advertisement or non-solicited unicast advertisement, S must be set to **0**. For example, during DAD, NS packets use the multicast address of the solicited node. When the addresses conflict with each other, S in the returned NA packet must be set to **0.**
- O: This field is the override flag. The value **1** indicates that a data-link-layer address among available destination data-link-layer addresses is used to update the neighbor cache table. The value **0** indicates that a target data-link-layer address can be used to update the neighbor cache table only when the data-link-layer address is unknown. In a solicited NA packet, if the destination address of the corresponding NS packet is an anycast address or the solicited agent advertisement, the override flag should not be set to **1**. In other cases, such as DAD, this flag should be set to **1**.
- Reserved: The sender must initialize the field value to **0**. The receiver must ignore this field.
- Target Address: This field is the destination address. If the NA message is sent in response to an NS message, the field value is the destination address of the NS message. If the NA message is not sent in response to an NS message, the field value is the IP address of the network node whose data-link-layer address has changed. The destination address cannot be a multicast address.
- Options: This field can only be set to a destination data-link-layer address, which is a data-link-layer address of the advertisement sender.

- Router solicitation (RS)

  After being started, the host sends an RS packet to the router, and the router responds with an RA packet. **Figure 4-17** shows the RS packet format.

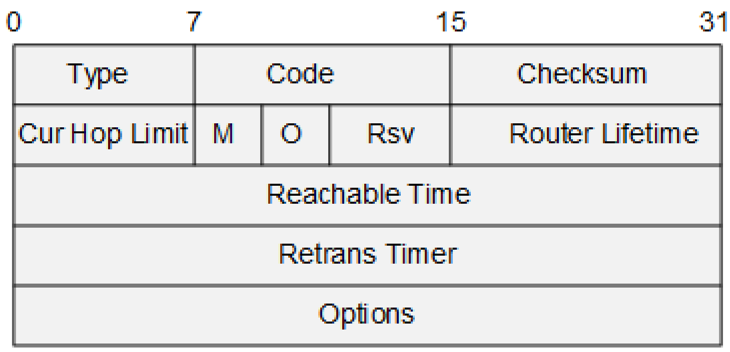
Figure 4-17 RS packet format

- Type: The value is **133**.
- Code: The value is **0**.
- Checksum: This is the checksum field.
- Reserved: The sender must initialize the field value to **0**. The receiver must ignore this field.
- Options: This field specifies the data-link-layer address of the packet sender. Only address options at the source data link layer can be used in RS packets. If the source address of an IPv6 header is an unassigned address, this field cannot be contained.

- Router advertisement (RA)

  A router sends RA packets, which include the prefix and flag bits, to respond to the RS packets sent by the host or RA packets periodically sent by the host. **Figure 4-18** shows the RA packet format.

Figure 4-18 RA packet format



- Type: The value is **134**.
- Code: The value is **0**.
- Checksum: This is the checksum field.
- Cur Hop Limit: This field specifies the maximum number of hops for packet sending of a router.
- M: This is the managed address configuration field. The value **0** indicates stateless address allocation. The host obtains the prefix in the prefix information option and automatically generates IPv6 addresses. The value **1** indicates stateful address allocation. The client obtains IPv6 addresses using the DHCPv6 protocol.
- Rsv: This is a reserved field. The sender must initialize the field value to **0**. The receiver must ignore this field.
- Router Lifetime: This field specifies the life cycle of the router that sends RA packets as the default router (unit: s).

- Reachable Time: This field specifies the reachable time (unit: ms). A router sends RA packets over an interface to enable all nodes on the same link to use the same reachable time.
- Retrans Timer: This field specifies the retransmission timer (unit: ms). This timer indicates the interval for retransmitting NS packets. It is used for unreachable neighbor detection and address resolution.
- Options: This is the option field.

NOTE

A node sends RA packets only if the node works in router mode and supports router forwarding. A node does not send RA packets as a host or a Layer 2 switch.

- Redirection packet

  IPv6 routers send redirection packets to redirect packets to a more suitable router. Redirection packets inform the host to select a more suitable next hop address. **Figure 4-19** shows the redirection packet format.

Figure 4-19 Redirection packet format

0	7	15	31
Type	Code	Checksum	
Reserved			
Target Address			
Destination Address			
Options			

- Type: The value is **137**.
- Code: The value is **0**.
- Checksum: This is the checksum field.
- Reserved: The sender must initialize the field value to **0**. The receiver must ignore this field.
- Target Address: This field specifies the target address, which is the next hop address of the destination address. If the destination is a router (packets outside the local link), the local link address of the router must be used. For a host (local-link packets), the target address and destination address must be the same.
- Destination Address: This field specifies the destination address, which is the destination address of the IPv6 packet header.
- Options: This is the option field.

4.4.3.2 Address Resolution

The address resolution function is similar to the Address Resolution Protocol (ARP) function of IPv4 transmission. The principles are as follows:

- When sending a service packet, a network node queries the neighbor cache table to obtain the link-layer address. If the link-layer address is not found, the network node sends an NS packet with the source IP address set to the link-local address of the NE and the destination IP address set to the multicast address of the solicited node. Multicast addresses can reduce the risk of network storms.

- When the solicited-node multicast address of the neighbor node on the link is the same as the destination IP address in the IP header of the NS packet and the destination address in the ICMP header is the same as the IP address of the local node, the NA packet is sent as a response. The neighbor discovery protocol on the node establishes a neighbor discovery entry according to information such as the link-layer address and IP address in the NS packet or the NA packet.
- When the link-layer address of a node changes, a non-solicited NA packet is sent to inform the neighbor on the link to update the neighbor cache entry.

On the base station side, run the **DSP ND** command to query the neighbor cache table.

4.4.3.3 Neighbor Unreachability Detection

The neighbor discovery protocol is used to detect whether a neighboring node that has established the neighbor relationship with a node is reachable. If the node receives an NA message in response to the NS message, the message sent to the neighboring node has been received by the neighboring node at the IP layer. In this case, the neighboring node is regarded as reachable. After a node determines the link-layer address of a neighboring node, the node traces the status of the neighbor cache entry and periodically sends an NS message. The destination IP address in the IP header of the message is a unicast address. If the neighboring node cannot respond with an NA message within the specified time, the neighbor cache entry is deleted.

Table 4-4 describes the five states of a neighbor cache entry.

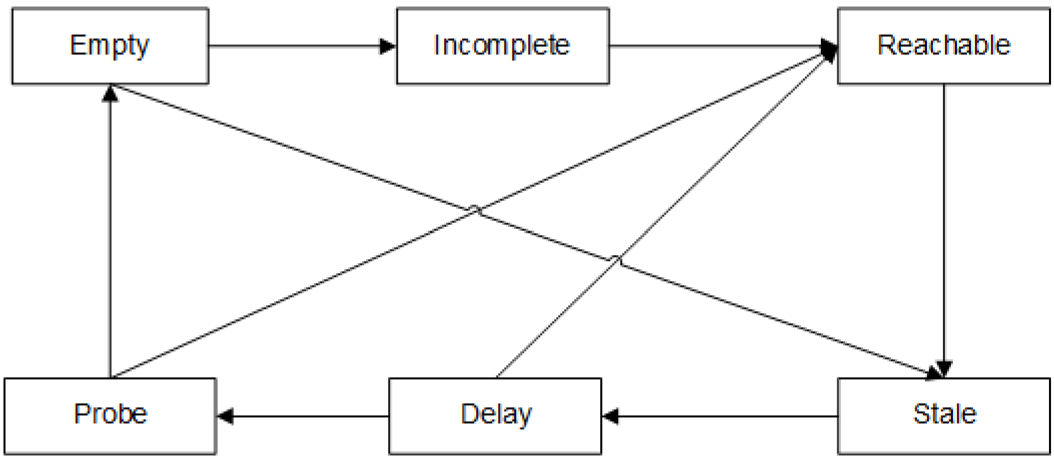
Table 4-4 Summary of neighbor cache entry states

State	Meaning
Incomplete	The destination IP address is being resolved. The link-layer address of the neighboring node is unknown. The NS message whose destination IP address is the solicited-node multicast address has been sent, but the NA response message from the peer node has not been received.
Reachable	The forwarding path of the neighboring node is normal, and the neighboring node is reachable within the neighbor discovery reachable time.
Stale	The neighboring node is no longer regarded as reachable. If the time in which the entry is in the Reachable state exceeds the neighbor discovery reachable time, the state changes to Stale. If no message is sent when the entry is in the Stale state, the state remains unchanged.

State	Meaning
Delay	The entry enters the Delay state if either of the following conditions is met: ● A message is sent when the entry is in the Stale state. ● The time when the entry is in the Stale state expires. After the entry enters the Delay state, the node sends an NS message. If the node receives an NA response message from the peer node, the entry changes to the Reachable state.
Probe	● In the Delay state, the node sends an NS message to check whether the peer node is reachable. If the node does not receive an NA response message from the peer node within the Delay_First_Probe_Time (5s), the entry changes to the Probe state. ● In the Probe state, the node sends NS messages for multiple times. If the node receives an NA response message from the peer node, the entry changes to the Reachable state. If no NA response message is received from the peer node, the entry is deleted.

For example, if node A needs to access node B and the neighbor cache entry of node A does not contain the entry of node B, the state of the neighbor cache entry changes as shown in **Figure 4-20**.

Figure 4-20 Neighbor discovery states



- Node A sends an NS message and generates a neighbor cache entry that is in the Incomplete state.
- If node B replies with an NA message, the neighbor cache entry changes from the Incomplete state to the Reachable state. Otherwise, the neighbor cache entry is deleted after 3s.

- In the Reachable state, if a non-request NA message is received from node B and the link-layer address is different before the neighbor discovery reachable time expires, the entry state changes to Stale immediately.
- In the Stale state, if node A needs to send data to node B, node A changes the entry state to the Delay state.
- In the Delay state, if no NA message is received within the Delay_First_Probe_Time (5s), the state changes to Probe. If an NA message is received, the state changes from Delay to Reachable.
- In the Probe state, a unicast NS message is sent every Retrans_Timer (1s). If a response is received after an NS message is sent for MAX_UNICAST_SOLICIT (three times), the entry state changes to Reachable. Otherwise, the entry is deleted.
- As defined in the RFC4861 standard, when a node receives other ND packets (such as RA and NS), it generates neighbor cache entries and is in the Stale state.

On the base station side, run the **SET ND** command with the **ND**.***NDREACHABLETIME (5G gNodeB, LTE eNodeB)*** parameter set to **30000** as the neighbor discovery reachable time in the Reachable state. The neighbor discovery reachable time must be planned together with the switch and router on the local link. If the value is too small, an excessive number of NS and NA messages will be generated on the network.

4.4.3.4 Duplicate Address Detection

Duplicate address detection (DAD) is a detection mechanism used to determine whether an IPv6 address is available. Neighbor discovery performs DAD before the unicast IPv6 address of an interface takes effect. The effective unicast address is known as a detection address. Neighbor discovery checks whether there are duplicate IP addresses on the network by sending an NS message to the detection address. The destination IP address in the NS message is set to the IP address to be detected. The source IP address in the IPv6 packet header is set to an unspecified address, and the destination IP address to the solicited-node multicast address. The DAD process is as follows:

- When a node is configured with or automatically generates an IPv6 unicast address, it immediately sends an NS message to check whether the IP address is available.
- After receiving the NS message, the neighbor node configured with the IPv6 address replies to the source node with an NA message carrying the IPv6 address.
- The source node receives the NA message from the neighboring node and considers that the IPv6 address is used by the neighboring node and is invalid. Conversely, if the source node does not receive any NA message from the neighboring node, the configured IPv6 address is available.
- If a link-local address conflict occurs, the response interface of the link-local address is inactive.

A base station performs DAD on the configured global unicast address and automatically generated link-local address.

NOTE

To enable DAD in special fault scenarios, the base station sends a small number of NS packets in the multicast domain after the unicast IPv6 address takes effect.

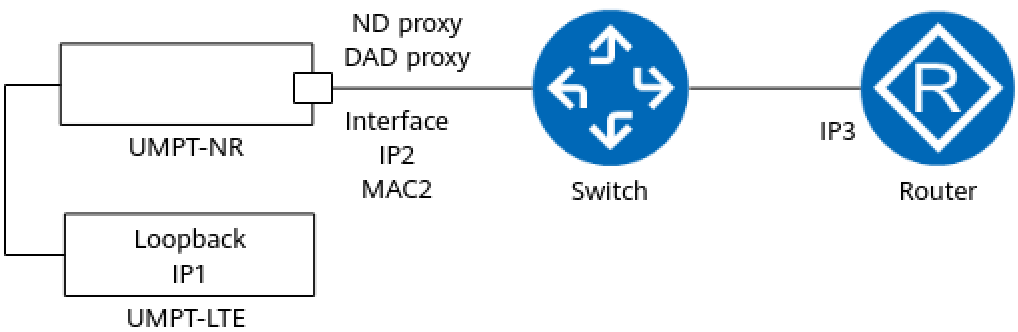
The link-local address generated based on the MAC address in EUI 64 format on the physical port of a base station is globally unique and does not conflict with IP addresses of other network devices. If the link-local addresses of other devices on the network are automatically generated using other methods or manually configured, they may conflict with the link-local address of the base station. When this occurs, the interface of the base station is unavailable. In this case, the link-local addresses of these network devices need to be changed. The link-local addresses of base stations cannot be configured manually. This prevents link-local address conflicts caused by incorrect configuration.

4.4.3.5 ND Proxy and DAD Proxy

The base station supports ND proxy and DAD proxy on an Ethernet port for IP addresses of different modes in backplane-based co-transmission scenarios. The Ethernet port acts as a proxy port and uses its own MAC address to establish the mapping between the proxy IP address and the MAC address with the router. In addition, the Ethernet port supports DAD for the proxy IP address.

As shown in **Figure 4-21**, the main control and transmission board of NR provides co-transmission through backplane interconnection with the main control and transmission board of LTE. IP2 is configured and ND proxy and DAD proxy are enabled on the Ethernet port of NR. Logical IP1 is configured on the LTE side. If IP1 is included in the network segment of IP2 and IP1 has taken effect, the Ethernet port can perform ND proxy for IP1, and the router can directly obtain the MAC address, MAC2, mapped from IP1. In addition, the Ethernet port sends NS packets for IP1 to detect address conflicts.

Figure 4-21 ND proxy and DAD proxy in backplane-based co-transmission scenarios



4.4.3.6 Router Discovery

In IPv6 transmission networking, the router discovery mechanism supports automatic address generation, default router discovery, and parameter discovery. This mechanism requires that routers be deployed on the transmission network.

This mechanism is implemented using RS and RA messages. The host sends an RS message and receives an RA response message during interface startup, or receives RA messages periodically sent by the router during normal operation. The RA message contains the following information:

- Address auto-configuration type
- Prefix information option, indicating the lifetime of the link-local address prefix
- Route information option
- Other information related to the host, such as the neighbor discovery reachable time and MTU that can be used by the message sent by the host
- Automatic address generation: If the auto-configuration type in an RA message is stateless auto-configuration, the host automatically generates an IPv6 address based on the prefix and local interface ID in the RA message.
- Default router discovery: The host sets the default router based on the default router information in the RA message.
- Parameter discovery: The host updates the interface parameters based on the parameter information in the RA message.

NOTE

Base stations do not support router discovery.

4.4.3.7 Route Redirection

A router notifies the originating node on the local link through a redirect message that a preferred neighboring router is available on the same link for packet forwarding. The node that receives the message modifies its local router entry accordingly. This function requires that routers be deployed on the transmission network.

Base stations do not support route redirection, and packets are forwarded based on static routes.

4.4.4 IP Layer Fragmentation

IPv6 has a different IP layer fragmentation mechanism from IPv4. In IPv6 transmission, the router does not support packet fragmentation. The router discards the packet whose length exceeds the MTU size. Then the router sends an ICMPv6 Packet Too Big message to the source NE to notify the source NE of the router's MTU capability. Only the source NE supports fragmentation of the packet whose length exceeds the MTU size and the destination NE reassembles the received fragments.

NOTE

A base station can fragment and reassemble only IP packets with a maximum of five fragments. Therefore, plan the MTU properly to prevent transmission exceptions of large-packet services.

4.4.4.1 Interface MTU

The IPv6 MTU can be configured on an interface of a base station through the **INTERFACE**.***MTU6 (5G gNodeB, LTE eNodeB)*** parameter.

It is recommended that the MTU of an interface be set to the minimum MTU on all backhaul networks of the interface.

By default, the base station complies with the RFC 8201 protocol and supports Path MTU Discovery. If the MTU is not set as recommended and the MTU of the interface is greater than the minimum MTU on all backhaul networks of the interface, the base station automatically adjusts the Path MTU of each link based on the ICMPv6 Packet Too Big message sent by the router. **Figure 4-22** shows the interface MTU configuration.

NOTE

Path MTU discovery of the base station requires that the router support the sending of ICMPv6 Packet Too Big messages and the network firewall allow such packets to pass through. If the L2+L3 networking is used, the maximum receive unit (MRU) of the L2 switch must be greater than the interface MTU of the base station. Otherwise, the router cannot send ICMPv6 Packet Too Big messages.

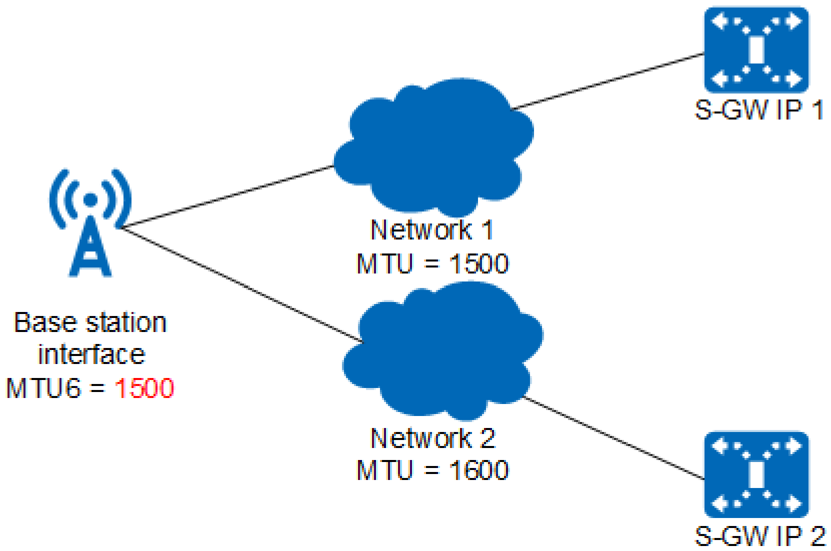
The default value of **PMTUCFG.*MODE (5G gNodeB, LTE eNodeB)*** is **PASSIVE** for the base station and the base station does not send discovery packets. If the base station receives the ICMPv6 Packet Too Big message from the router, this indicates that the router has discarded a small number of service packets.

In passive mode, if the MTU size of the actual outbound interface (**INTERFACE.*MTU6***) increases, the Path MTU entry will be updated the next time.

The maximum IPv6 receive unit before reassembly is limited over the Ethernet port on the base station. The maximum IPv6 receive unit among boards on the base station is equal to the maximum IPv6 MTU of the base station. For details, see the notes in the **ADD INTERFACE** command help.

To prevent packet loss in the receive direction of the base station because the packet size exceeds the maximum IPv6 receive unit, you are advised to properly plan the MTUs of the base station and router.

Figure 4-22 Configuring an interface MTU

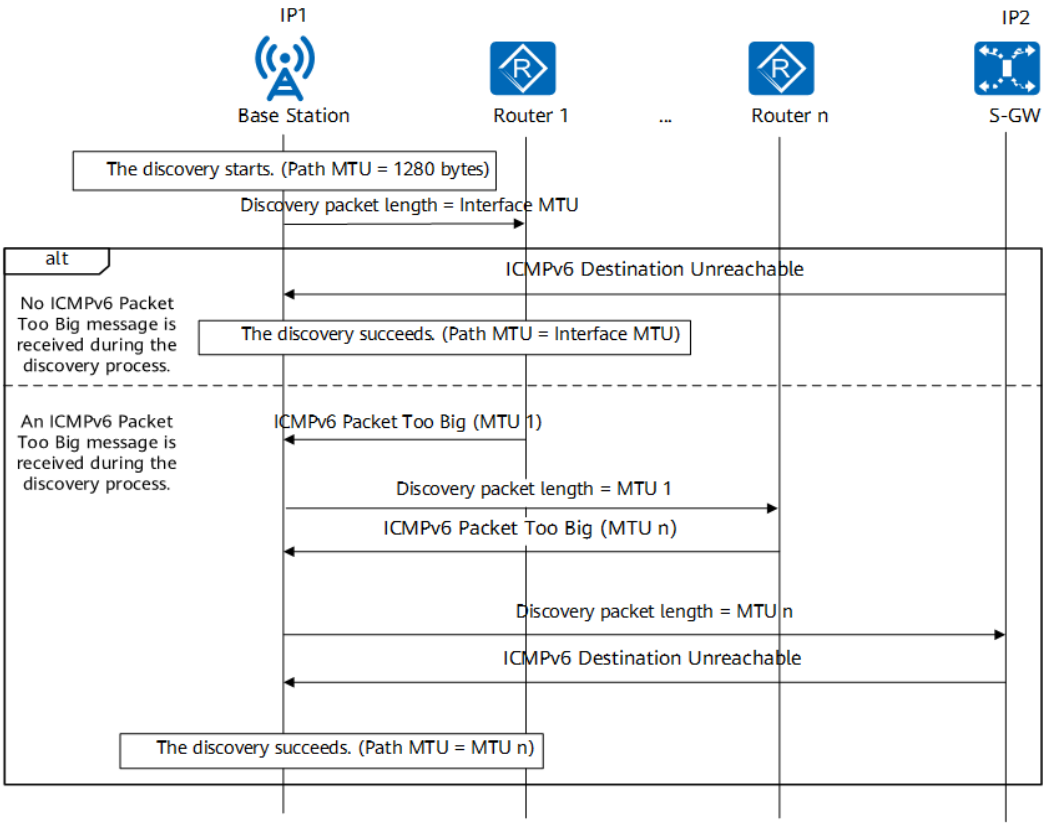


4.4.4.2 Path MTU Discovery

The base station supports path MTU discovery when **PMTUCFG.*MODE (5G gNodeB, LTE eNodeB)*** is set to **ACTIVE**. The base station actively sends packets for path MTU discovery on each service link. If the MTU is not set as recommended, this function can adjust the path MTU of each link in time.

Figure 4-23 shows the path MTU discovery process.

Figure 4-23 Path MTU discovery process



1. The base station initiates the MTU discovery to the destination node on each service link. The initial length of a discovery packet is the interface MTU size. During the discovery, the path MTU size is 1280 bytes. For an IKE link, if **IKEPEER**.***IPSECPREFRGSW (5G gNodeB, LTE eNodeB)*** is set to **OFF** before encryption, the first detection result is the Path MTU in passive mode after the passive mode is switched to the active mode.
2. If an ICMPv6 Packet Too Big message is received within the period specified by **PMTUCFG**.***TIMEOUTDUR (5G gNodeB, LTE eNodeB)***, this indicates that the length of the detection packet exceeds the MTU of the router. After recording the MTU of the router, the base station adjusts the length of the detection packet and initiates the detection again.
3. If another ICMPv6 Packet Too Big message is received within the period specified by **PMTUCFG**.***TIMEOUTDUR (5G gNodeB, LTE eNodeB)***, repeat step 2. If an ICMPv6 Destination Unreachable message is received within the period specified by **PMTUCFG**.***TIMEOUTDUR (5G gNodeB, LTE eNodeB)***, this indicates that the discovery is successful.

Successful Discovery

After the discovery is successful, the base station uses the router MTU that is last recorded as the path MTU of the link. If the base station does not receive an ICMPv6 Packet Too Big message during the discovery, the base station changes the path MTU to the interface MTU.

Abnormal Discovery

In the discovery process, the following are true within the period specified by **PMTUCFG**.***TIMEOUTDUR (5G gNodeB, LTE eNodeB)***:

- If neither the ICMPv6 Packet Too Big nor ICMPv6 Destination Unreachable message is received, the Path MTU of the link is changed to the interface MTU.
- If the ICMPv6 Packet Too Big message is received but the ICMPv6 Destination Unreachable message is not, the Path MTU of the link is changed to the router MTU recorded last time.

Aging and Update

- After the discovery is successful, set the **PMTUCFG**.***AGINGTIME (5G gNodeB, LTE eNodeB)*** parameter to specify the aging time of the path MTU. After the path MTU is aged out, if the service data of a link is sent within 4 hours, the discovery is started again. During the discovery, the path MTU is the value before the aging. If no service data is sent over a link within 4 hours, the path MTU value before the aging is cleared. When there is service data to be sent, the discovery is started again, which follows the same process.
- If the discovery is abnormal, the base station automatically uses the default path MTU discovery mechanism after the path MTU is aged out.

NOTE

If the following deployment conditions of path MTU discovery are not met, path MTU discovery will be abnormal:

- The router must support the sending of ICMPv6 Packet Too Big messages.
- The peer NE must support the sending of ICMPv6 Destination Unreachable messages, and the destination port of the discovery (specified by **PMTUCFG**.***PORTNO (5G gNodeB, LTE eNodeB)***) on the base station must be a non-service port of the peer NE.
- The network firewall must allow the preceding ICMPv6 packets and discovery packets of the base station to pass through.

UDP packets are sent during path MTU discovery. If there are a large number of links in a base station, the peer NE will receive a large number of discovery packets.

During path MTU discovery, if the route of the base station or the MTU of the router is adjusted, the new MTU fails to be discovered in some scenarios. The MTU can be obtained after the next discovery is complete.

During a base station upgrade, the first discovered MTU may be inaccurate due to the sequence in which the base station routes take effect. The MTU can be updated after the next discovery is complete.

4.5 Transport Layer

The IPv6 transport layer consists of four protocols: User Datagram Protocol (UDP), Stream Control Transmission Protocol (SCTP), Transmission Control Protocol (TCP), and GPRS Tunneling Protocol-User Plane (GTP-U). The protocol functions are the same as those of IPv4. For details, see *IPv4 Transmission*.

- UDP is used to transmit data. UDP is a connectionless and unreliable transport layer protocol based on the IP layer.
- SCTP runs over the IP layer and provides reliable connection-oriented transmission between two SCTP endpoints by establishing an SCTP

association. For details, see RFC 4960. Only one SCTP association is established between two endpoints.

- TCP is a reliable connection-oriented transport layer protocol. It is used for carrying OM channels of LTE and NR.
- GTP-U is used to transmit user data of the S1 and X2 interfaces of LTE and NR. It complies with 3GPP TS 29.281 and runs over UDP.

4.6 IPv6 Transport Protocol Stacks for Interfaces

4.6.1 Protocol Stacks for LTE Transport Interfaces

Transmission interfaces on the E-UTRAN include the S1 and X2/eX2 interfaces. Control- and user-plane protocols for these interfaces are essential to the communications between an eNodeB and its peer devices and support IPv6 transmission.

NB-IoT supports only S1 and X2-C (control plane) interfaces and supports IPv6 transmission.

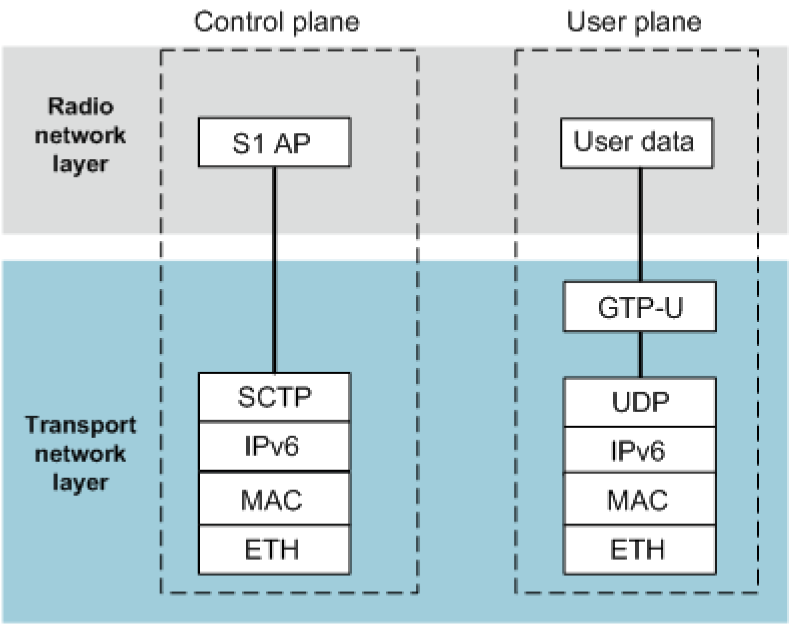
The transport bearer protocols above the IP layer in the following interface protocol stacks are the same as those in IPv4 transmission. For details, see *IPv4 Transmission*.

S1 Interface

As shown in **Figure 4-24**, the control plane and user plane of the S1 interface use IPv6, and the data link layer and physical layer use Ethernet.

- The control plane of the S1 interface uses SCTP at the transport layer.
- The user plane of the S1 interface uses GTP-U over UDP/IP at the transport layer.

Figure 4-24 Protocol stack for the S1 interface

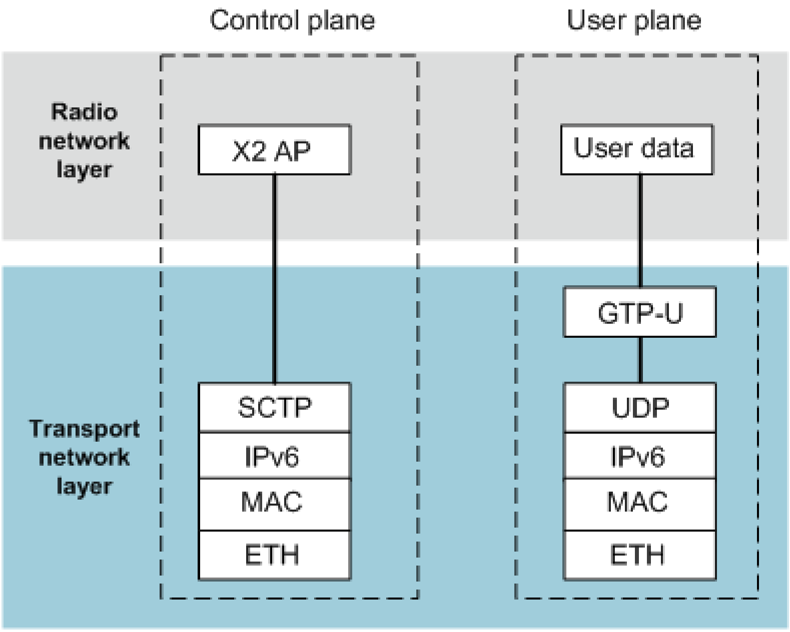


X2 Interface

As shown in **Figure 4-25**, the control plane and user plane of the X2 interface use IPv6, and the data link layer and physical layer use Ethernet.

- The control plane of the X2 interface uses SCTP at the transport layer.
- The user plane of the X2 interface uses GTP-U over UDP/IP at the transport layer.

Figure 4-25 Protocol stack for the X2 interface

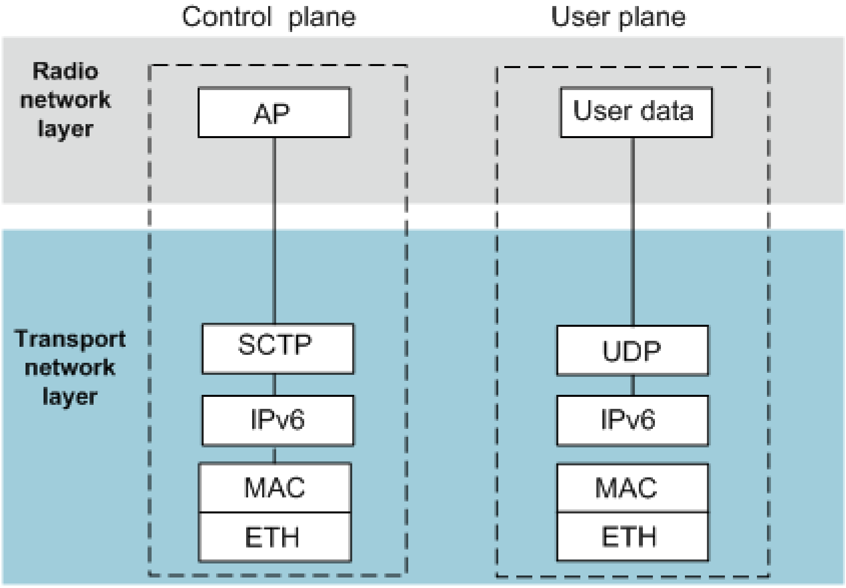


eX2 Interface

As shown in **Figure 4-26**, the control plane and user plane of the eX2 interface use IPv6, and the data link layer and physical layer use Ethernet.

- The control plane of the eX2 interface uses SCTP at the transport layer.
- The user plane of the eX2 interface uses UDP/IP at the transport layer.

Figure 4-26 Protocol stack for the eX2 interface



4.6.2 Protocol Stacks for NSA NR Transport Interfaces

Transmission interfaces on the NSA NR RAN include the S1-U, X2, and eXn-U interfaces. Control- and user-plane protocols for these interfaces are essential to the communications between a gNodeB and its peer devices.

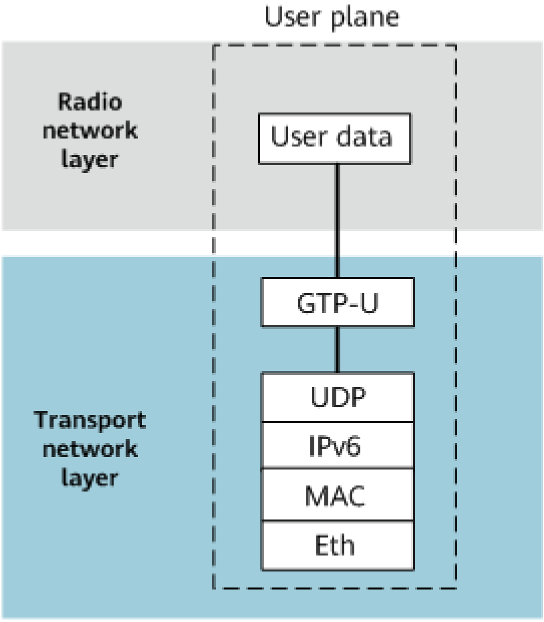
The transport bearer protocols above the IP layer in the following interface protocol stacks are the same as those in IPv4 transmission. For details, see *IPv4 Transmission*.

S1 Interface

Figure 4-27 shows the protocol stack for the S1 interface. The transmission bearer protocols above the IP layer are the same as those in IPv4 transmission. The user plane uses IPv6, and the data link layer and physical layer use Ethernet.

The user plane of the S1 interface uses GTP-U over UDP/IP at the transport layer.

Figure 4-27 Protocol stack for the S1 interface user plane

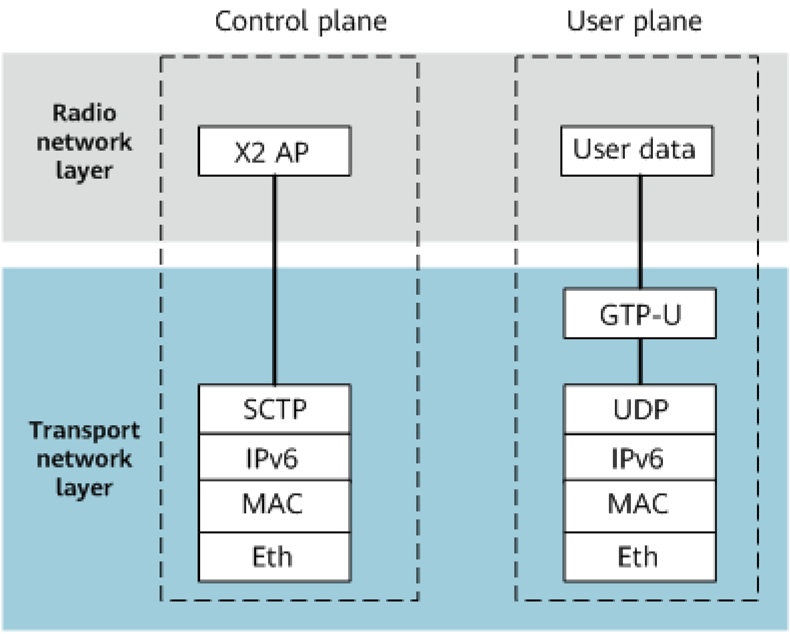


X2 Interface

As shown in **Figure 4-28**, the control plane and user plane of the X2 interface use IP, and the data link layer and physical layer use Ethernet.

- The control plane of the X2 interface uses SCTP at the transport layer.
- The user plane of the X2 interface uses GTP-U over UDP/IP at the transport layer.

Figure 4-28 Protocol stack for the X2 interface

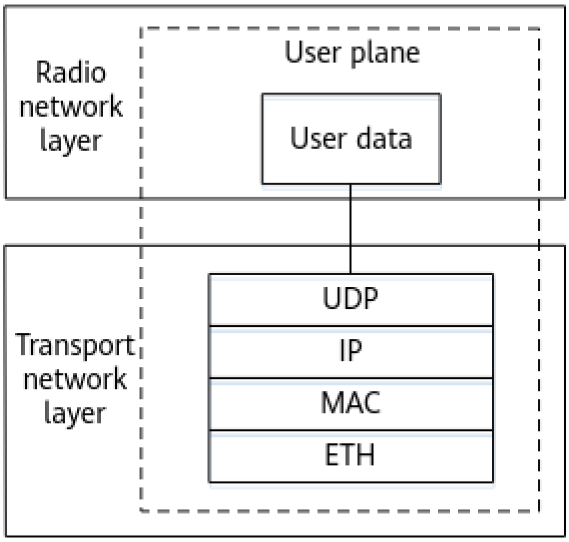


eXn Interface

As shown in **Figure 4-29**, the user plane of the eXn interface uses IP, and the data link layer and physical layer use Ethernet.

The user plane of the eXn interface uses UDP/IP at the transport layer.

Figure 4-29 Protocol stack for the eXn interface



4.6.3 Protocol Stacks for SA NR Transport Interfaces

Transmission interfaces on the 5G RAN include the NG, Xn, and eXn-U interfaces. Control- and user-plane protocols for these interfaces are essential to the communications between a gNodeB and its peer devices. IPv6 transmission is supported.

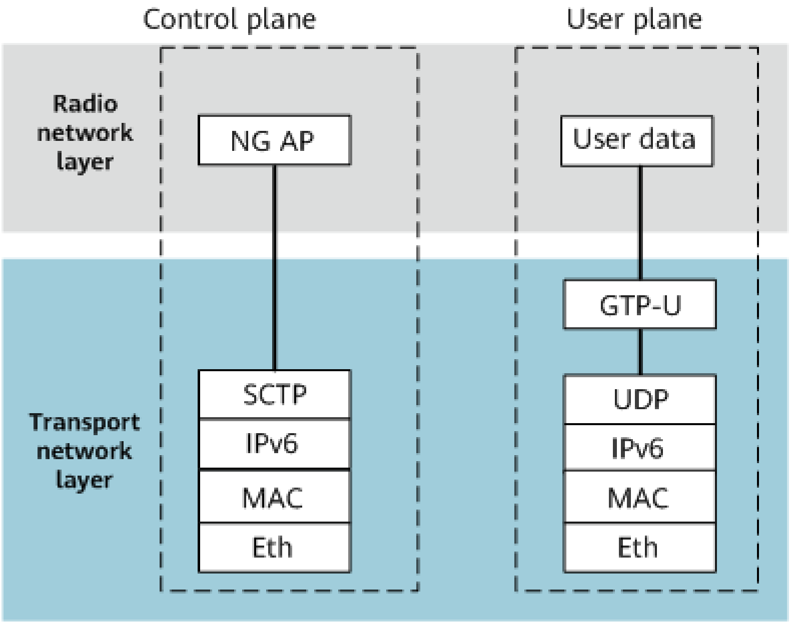
The transport bearer protocols above the IP layer in the following interface protocol stacks are the same as those in IPv4 transmission. For details, see *IPv4 Transmission*.

NG Interface

As shown in **Figure 4-30**, the control plane and user plane of the NG interface use IPv6, and the data link layer and physical layer use Ethernet.

- The control plane of the NG interface uses SCTP at the transport layer.
- The user plane of the NG interface uses GTP-U over UDP/IP at the transport layer.

Figure 4-30 Protocol stack for the NG interface

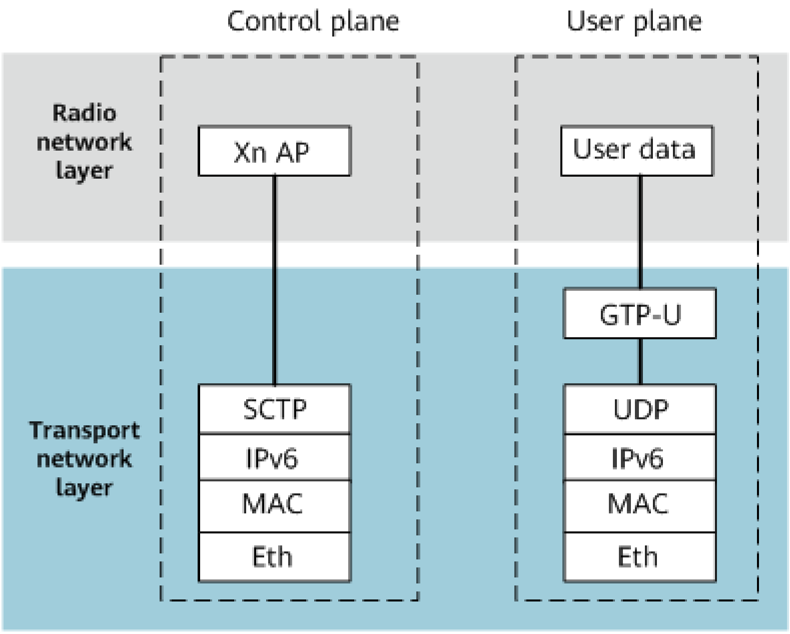


Xn Interface

As shown in **Figure 4-31**, the control plane and user plane of the Xn interface use IPv6, and the data link layer and physical layer use Ethernet.

- The control plane of the Xn interface uses SCTP at the transport layer.
- The user plane of the Xn interface uses GTP-U over UDP/IP at the transport layer.

Figure 4-31 Protocol stack for the Xn interface

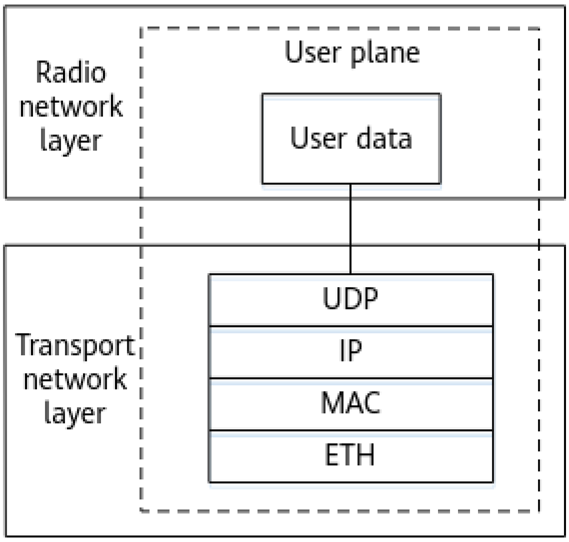


eXn Interface

As shown in **Figure 4-32**, the user plane of the eXn interface uses IP, and the data link layer and physical layer use Ethernet.

The user plane of the eXn interface uses UDP/IP at the transport layer.

Figure 4-32 Protocol stack for the eXn interface



4.6.4 Protocol Stacks for Other Transport Interfaces

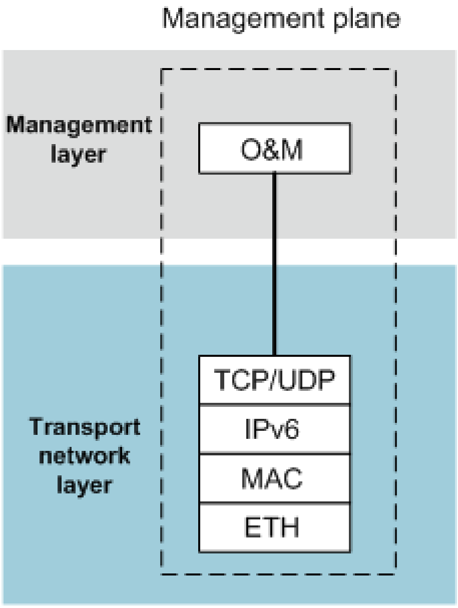
In addition to LTE and NR service interfaces, IPv6 transmission supports the management-plane interface and IP clock synchronization interface for wireless devices.

OM

The management-plane interface is the interface between the OSS and the base station.

The management-plane interface mainly uses the TCP over IP protocol stack. In addition to the TCP protocol, some protocols use UDP over IP. For example, NTP. The OM channel for the base station uses IPv6 transmission over FE or GE ports.

Figure 4-33 OM channel protocol stack



OM channel IPv6 transmission can be configured by specifying parameters in the **OMCH** MO. The **OMCH**.***BEAR (LTE eNodeB, 5G gNodeB)*** parameter is set to **IPV6**, and the **OMCH**.***IP6 (LTE eNodeB, 5G gNodeB)*** parameter can be set to an IPv6 address shared with other interfaces or to an independent IPv6 address.

The local IPv6 address of the OMCH can be configured using either of the following methods:

- Configure the **IPADDR6** MO, and then set the **OMCH**.***IP6 (LTE eNodeB, 5G gNodeB)*** parameter to the same value as that of the **IPADDR6**.***IPV6 (LTE eNodeB, 5G gNodeB)*** parameter. This method can be used when the IPv6 address of the OMCH needs to be set to an interface IP address.
- Set the **OMCH**.***IP6 (LTE eNodeB, 5G gNodeB)*** parameter in the **OMCH** MO. You do not need to configure the **IPADDR6** MO. This IP address is the IP address of the loopback interface. This method is used when main control boards work in active/standby mode.

For both configuration methods, ensure that the IPv6 route between the local IPv6 address of the OMCH and the MAE is reachable.

NOTE

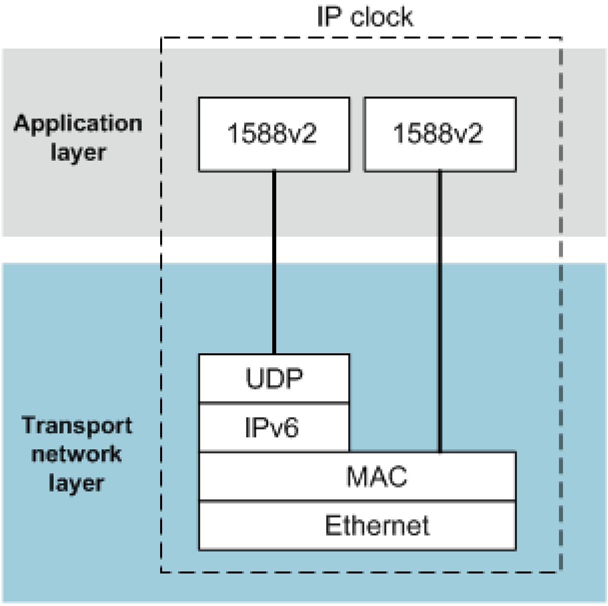
The OMCH can be established only when the remote maintenance IP address of the base station configured in the MAE topology view is the same as the local IPv6 address of the OMCH.

IP Clock Interface

The IP clock interface is used to provide 1588v2-based clock synchronization for base stations through IPv6 packets. For more information, see *Synchronization*.

The IP layer of the 1588v2 Layer 3 unicast protocol stack uses IPv6, and Layer 2 multicast does not involve the IP layer, as shown in **Figure 4-34**.

Figure 4-34 Protocol stack for the IP clock interface



4.6.5 Restrictions

- Base station models

  BTS3900, BTS3900 LTE, BTS5900, BTS5900 LTE, and BTS5900 5G
- Board type

  Only the UMPT and UMDU boards support this function.
- Base station software version

  A base station of SRAN15.1 or later is required in the following scenarios:
  - The base station is configured with IPv6 or dual-stack transmission.
  - The base station is configured with only IPv4 transmission but the peer device (base station or core network device) is configured with dual-stack transmission. The peer device sends a dual-stack message to the base station for interface self-setup.
- Base station networking

  IPv6 transmission does not support the following networking scenarios:
  - The base station is configured with multiple main control boards. For example, two UMPT boards are configured for transmission load sharing, or one UMPT is used to transmit data and the other is used as a signaling extension board.
  - The base station is configured with extension transmission boards to transmit data.
  - IPv6 transmission is implemented through inter-BBU interconnection.
  - The base station implements IPv6 co-transmission through the backplane.
- VLAN planning and configuration

  In dual-stack transmission, both IPv4 and IPv6 must be configured with VLANs based on interfaces.
- Interconnection

The IPv6 packets sent by the peer device can carry only the following extension headers. Otherwise, the packets will be discarded by the base station.

- Hop-by-hop options header: This header can be carried only in packets compliant with the MLD snooping discovery protocol.
- Fragment header: This header is used for packet fragmentation.
- Authentication header: This header is used to ensure IP security.
- ESP header: This header is used to ensure IP security.

5 IPv6 Transmission Networking for Interfaces

5.1 Overview

This chapter describes IPv6 transmission networking for the interfaces of LTE and NR.

5.2 LTE Interface Networking

On a live network, the eNodeB generally connects to other NEs, such as the S-GW, MME, eNodeB, IP clock server, and MAE, through a transport bearer network (backhaul). LTE IPv6 transmission supports Layer 2 and Layer 3 networking.

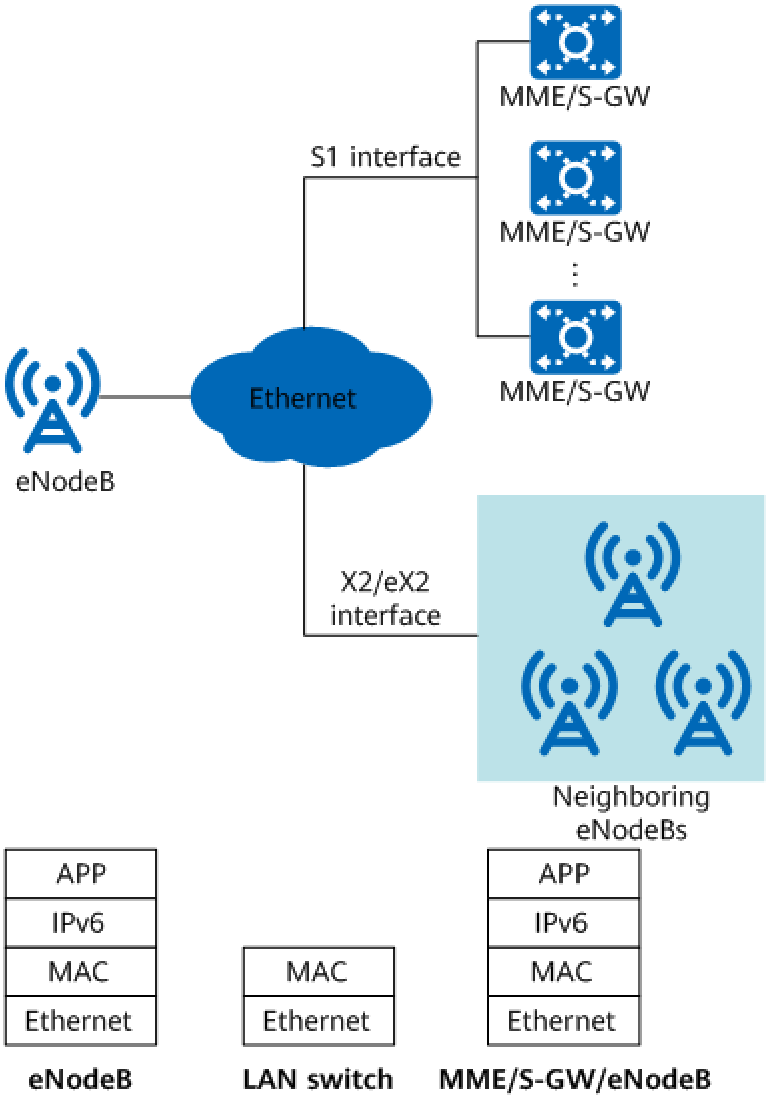
NOTE

In LTE networking, only the main control board of the eNodeB can provide IPv6 transmission.

Layer 2 Networking

IPv6 transmission supports Layer 2 networking. The Layer 2 network in the LTE system is the Ethernet switch network, which consists of Ethernet switches. The eNodeB accesses this network through the FE/GE/10GE port. There is no router on the Layer 2 network.

Figure 5-1 Layer 2 network



As shown in **Figure 5-1**, a Layer 2 network provides the bearer function at the MAC layer. An eNodeB does not provide the Layer 2 Ethernet switching function, and supports only packet forwarding based on IPv6 addresses. After the eNodeB encapsulates packets into Ethernet frames, the Layer 2 switch forwards Ethernet frames based on the source and destination MAC addresses.

You need to configure the Ethernet port, MAC layer, and IP layer, as described in **Table 5-1**.

Table 5-1 Layer 2 network configuration items

Configuration Item	MO	Configuration Description	Interface
Ethernet	**ETHPORT**	Ethernet negotiation parameters, such as the duplex mode and rate, must be set. Automatic negotiation is used by default.	FE/GE/10GE

Configuration Item	MO	Configuration Description	Interface
MAC layer	**INTERFACE**	Interfaces, IEEE 802.1p/q, and MTUs must be configured.	-
IP layer	**IPADDR6**	IPv6 addresses must be configured. Services must use the global unicast address instead of the link-local address.	IPv6 address

Layer 3 Networking

A Layer 3 network is an IP route-based switching network and mainly consists of routers. The eNodeB accesses the Layer 3 network through the FE/GE/10GE port.

Figure 5-2 Layer 3 network

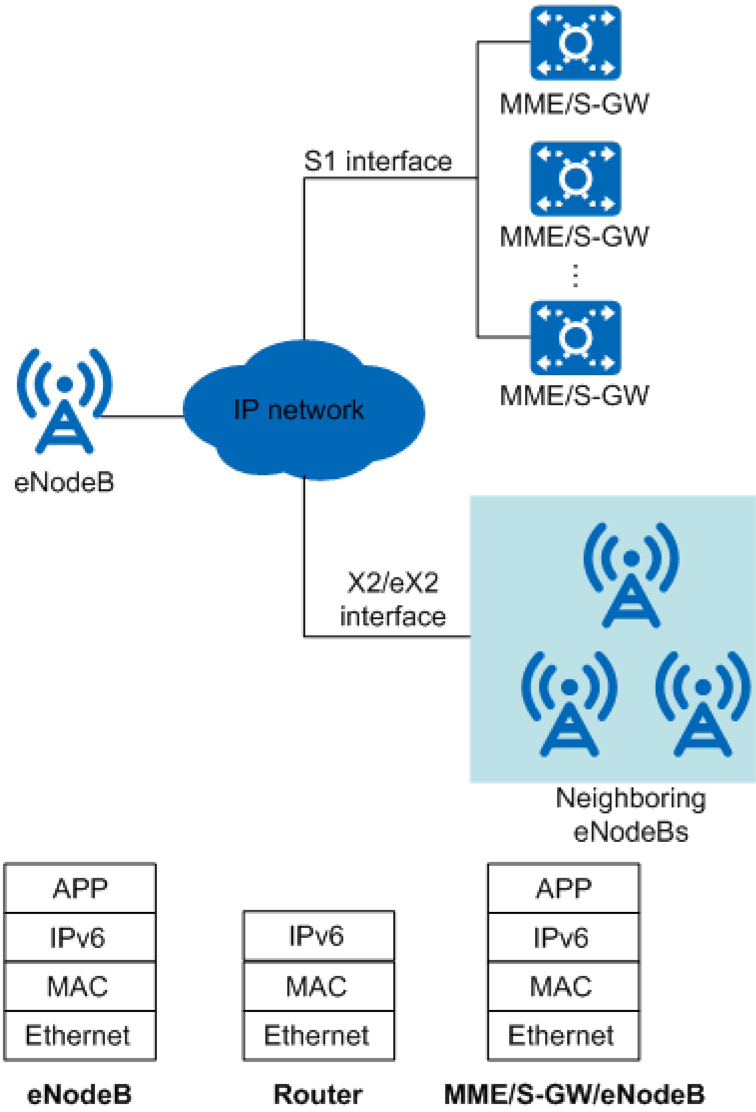


Figure 5-2 shows a Layer 3 network that provides the bearer function at the IPv6 layer, which is commonly applied to LTE networks. Users must configure the physical layer, data link layer, and IP layer. **Table 5-2** describes the configurations of a Layer 3 network over the FE/GE/10GE port.

Table 5-2 Configuration of a Layer 3 network over the FE/GE/10GE port

Configuration Item	MO	Configuration Description	Interface
Physical layer	**ETHPORT**	The configuration items are the same as those on a Layer 2 network. For details, see **Table 5-1**.	FE/GE/10GE
Data link layer	**INTERFACE**	The configuration items are the same as those on a Layer 2 network. For details, see **Table 5-1**.	None
IP layer	**IPADDR6** and **IPROUTE6**	IPv6 addresses, IPv6 routing tables, and DiffServ values must be configured.	IPv6 address

5.3 NSA NR Interface Networking

On a live network, the gNodeB generally connects to other NEs, such as the S-GW, eNodeB, IP clock server, and MAE, through a transport bearer network (backhaul). In IP-based networking, transport bearer networks are classified into Layer 2 and Layer 3 networks.

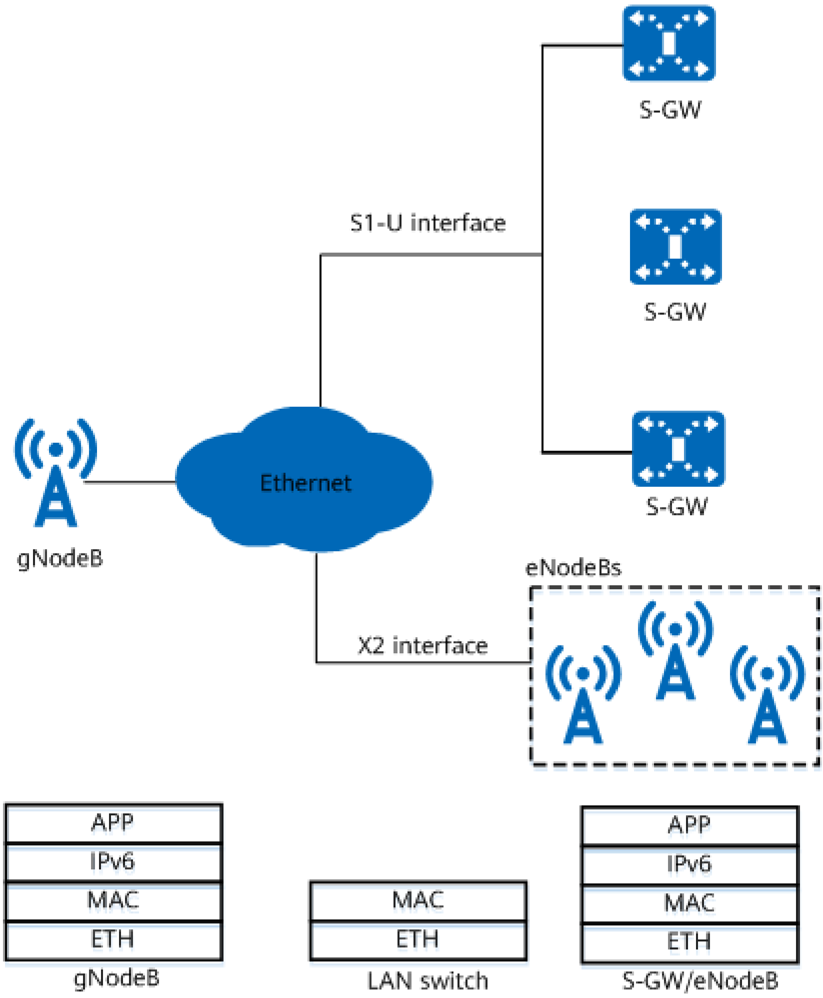
NOTE

In NR NSA networking, only the main control board of the gNodeB can provide IPv6 transmission.

Layer 2 Networking

A Layer 2 network is an Ethernet switching network and mainly consists of Ethernet switches. The gNodeB accesses this network through the FE/GE/10GE port.

Figure 5-3 Layer 2 network



As shown in **Figure 5-3**, a Layer 2 network provides the bearer function at the MAC layer. A gNodeB does not provide the Layer 2 Ethernet switching function, and supports only packet forwarding based on IPv6 addresses. After the gNodeB encapsulates packets into Ethernet frames, the Layer 2 switch forwards Ethernet frames based on the source and destination MAC addresses.

You need to configure the Ethernet port, data link layer, and IP layer, as described in **Table 5-3**.

Table 5-3 Layer 2 network configuration items

Configuration Item	MO	Configuration Description
Ethernet port	**ETHPORT**	Ethernet negotiation parameters, such as the duplex mode and rate, must be set. Automatic negotiation is used by default.
Data link layer	**INTERFACE**	Interfaces and IEEE 802.1p/q must be configured.

Configuration Item	MO	Configuration Description
IP layer	**IPADDR6**	IPv6 addresses must be configured.

Layer 3 Networking

A Layer 3 network is an IP route-based switching network and mainly consists of routers. The gNodeB accesses this network through the FE/GE/10GE port.

Figure 5-4 Layer 3 network

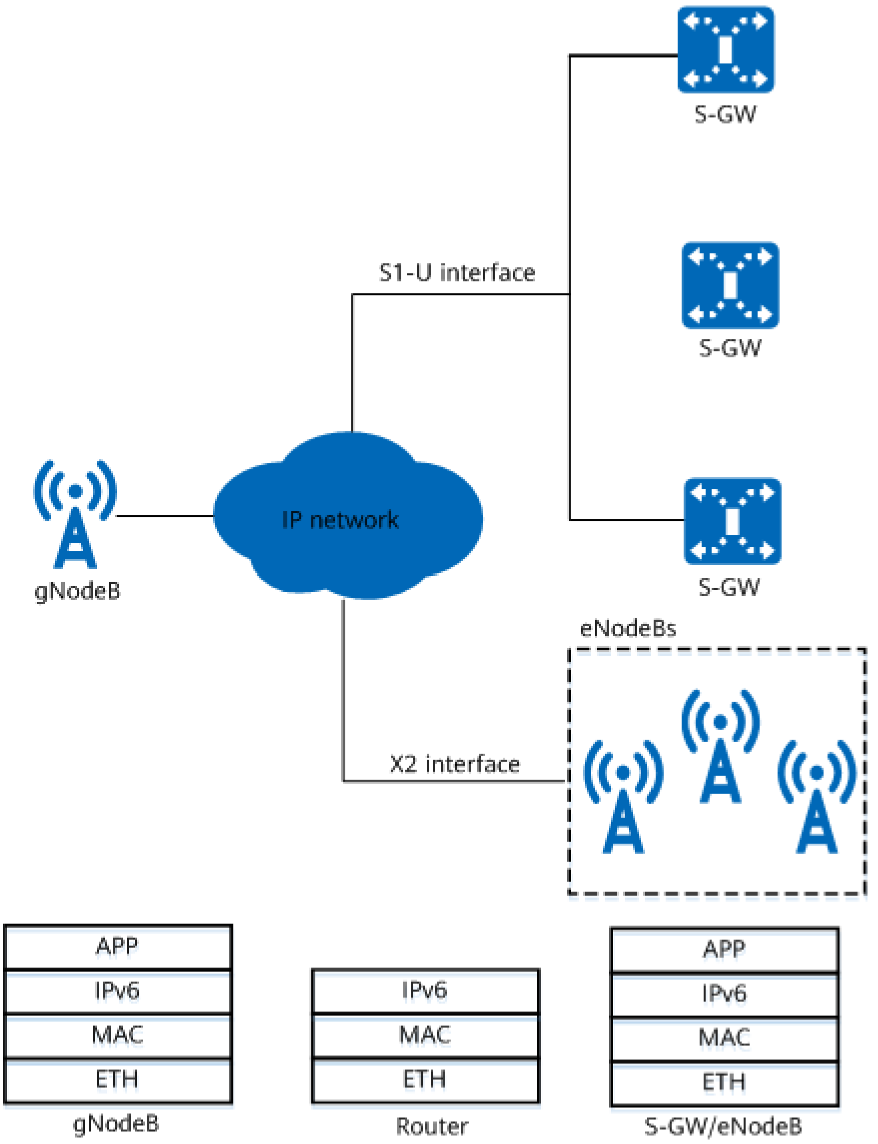


Figure 5-4 shows a Layer 3 network that provides the bearer function at the IPv6 layer, which is commonly applied to NR networks. Users must configure the physical layer, data link layer, and IP layer. **Table 5-4** describes the configurations of a Layer 3 network over the FE/GE/10GE port.

Table 5-4 Layer 3 network configuration items

Configuration Item	MO	Configuration Description
Physical layer	**ETHPORT**	The configuration items are the same as those on a Layer 2 network. For details, see **Table 5-3**.
Data link layer	**INTERFACE**	The configuration items are the same as those on a Layer 2 network. For details, see **Table 5-3**.
IP layer	**IPADDR6** and **IPROUTE6**	IPv6 addresses, IPv6 routing tables, and DiffServ values must be configured.

5.4 SA NR Interface Networking

On a live network, the gNodeB generally connects to other NEs, such as the 5GC, gNodeB, IP clock server, and MAE, through a transport bearer network (backhaul). In IP-based networking, transport bearer networks are classified into Layer 2 and Layer 3 networks.

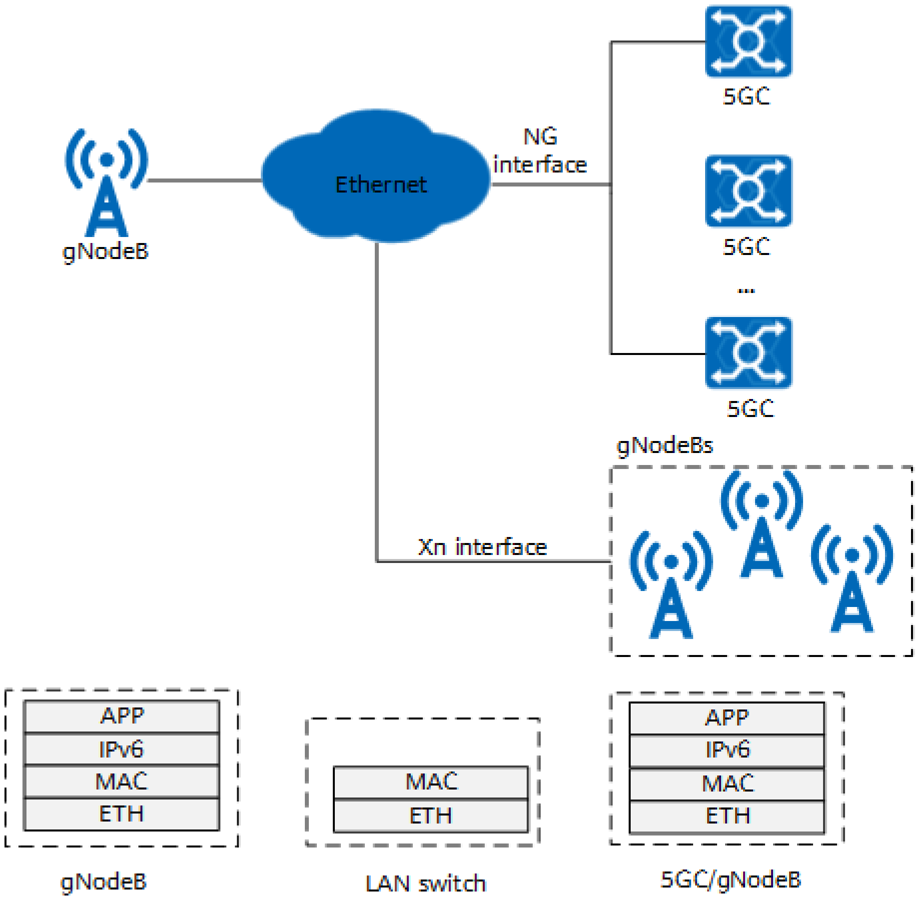
NOTE

In NR SA networking, only the main control board of the gNodeB can provide IPv6 transmission.

Layer 2 Networking

A Layer 2 network is an Ethernet switching network and mainly consists of Ethernet switches. The gNodeB accesses this network through the FE/GE/10GE port.

Figure 5-5 Layer 2 network



As shown in **Figure 5-5**, a Layer 2 network provides the bearer function at the MAC layer. A gNodeB does not provide the Layer 2 Ethernet switching function, and supports only packet forwarding based on IPv6 addresses. After the gNodeB encapsulates packets into Ethernet frames, the Layer 2 switch forwards Ethernet frames based on the source and destination MAC addresses.

You need to configure the Ethernet port, data link layer, and IP layer, as described in **Table 5-5**.

Table 5-5 Layer 2 network configuration items

Configuration Item	MO	Configuration Description
Ethernet port	**ETHPORT**	Ethernet negotiation parameters, such as the duplex mode and rate, must be set. Automatic negotiation is used by default.
Data link layer	**INTERFACE**	Interfaces and IEEE 802.1p/q must be configured.
IP layer	**IPADDR6**	IPv6 addresses must be configured.

Layer 3 Networking

A Layer 3 network is an IP route-based switching network and mainly consists of routers. The gNodeB accesses this network through the FE/GE/10GE port.

Figure 5-6 Layer 3 network

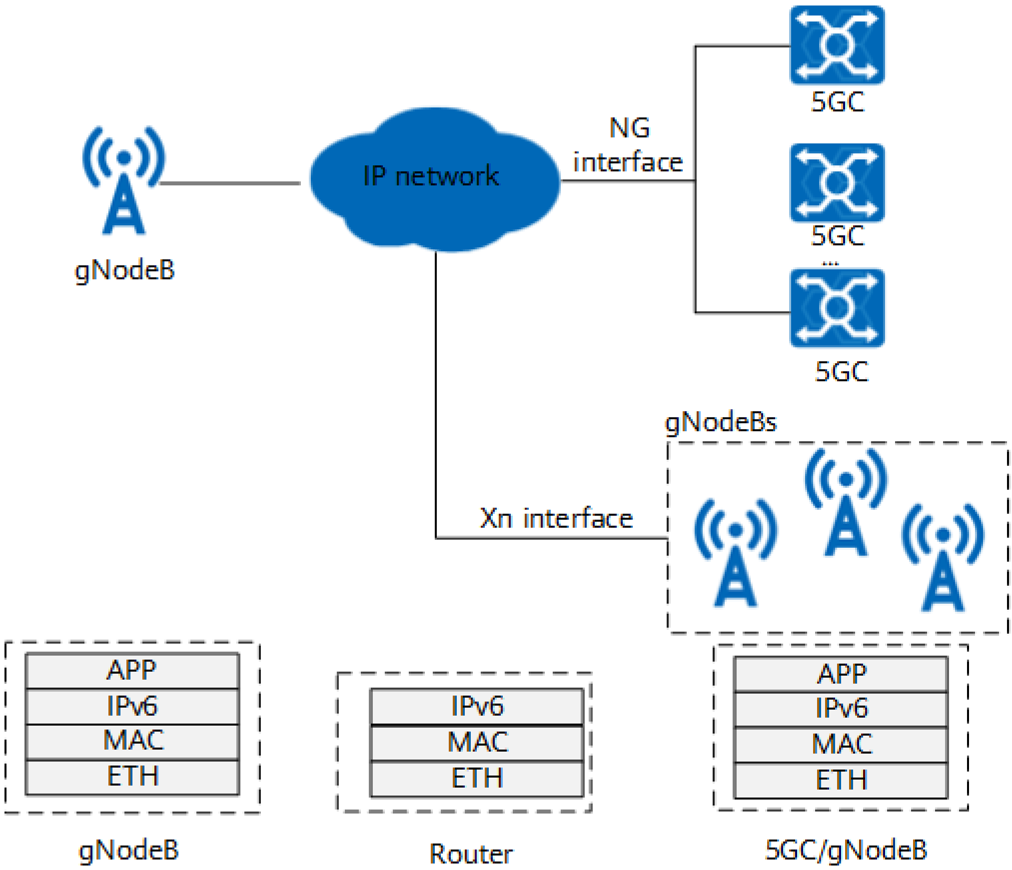


Figure 5-6 shows a Layer 3 network that provides the bearer function at the IPv6 layer, which is commonly applied to NR networks. Users must configure the physical layer, data link layer, and IP layer. **Table 5-6** describes the configurations of a Layer 3 network over the FE/GE/10GE port.

Table 5-6 Layer 3 network configuration items

Configuration Item	MO	Configuration Description
Physical layer	**ETHPORT**	The configuration items are the same as those on a Layer 2 network. For details, see **Table 5-5**.
Data link layer	**INTERFACE**	The configuration items are the same as those on a Layer 2 network. For details, see **Table 5-5**.

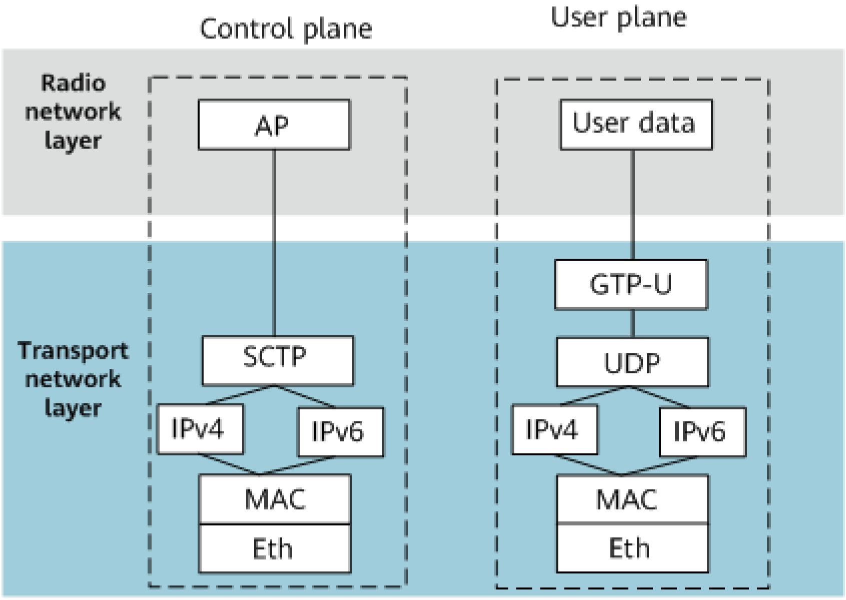
Configuration Item	MO	Configuration Description
IP layer	**IPADDR6** and **IPROUTE6**	IPv6 addresses, IPv6 routing tables, and DiffServ values must be configured.

6 IPv4/IPv6 Dual-Stack Networking for Interfaces

6.1 Overview

As shown in **Figure 6-1**, a dual-stack node can be configured with both IPv4 and IPv6 protocol stacks. The transport layer protocols, such as TCP, UDP, and SCTP, are the same for IPv4 and IPv6. Dual-stack nodes can communicate with both nodes that support IPv4 and those supporting IPv6.

Figure 6-1 IPv4/IPv6 dual stack



An eNodeB or gNodeB can be configured with only one type of protocols to work as a single-stack node, or be configured with both IPv4 and IPv6 protocols to work as a dual-stack node.

- IPv6 single-stack node

When only IPv6 transmission is used between the base station and the peer device, only the user plane and control plane complying with the IPv6 protocol are configured on the base station.

- IPv4 single-stack node

  When only IPv4 transmission is used between the base station and the peer device, only the user plane and control plane complying with the IPv4 protocol are configured on the base station.

- IPv4/IPv6 dual-stack node

  When IPv4 transmission is used between the base station and some peer devices and IPv6 transmission is used between the base station and the other peer devices, the user plane and control plane complying with the IPv4 and IPv6 protocols need to be configured on the base station.

  The IP protocol version of the base station must be consistent with that of the peer device. Otherwise, the link cannot be set up.

Table 6-1 Summary of IP protocol configurations between the base station and the peer device

IP Protocol Version for the eNodeB or gNodeB	IP Protocol Version for the Peer Device (Such as the Core Network and eNodeB/gNodeB)	IP Protocol Version for Link Setup
IPv4	IPv4	IPv4
IPv6	IPv6	IPv6
IPv4/IPv6	IPv4	IPv4
IPv4/IPv6	IPv6	IPv6
IPv4	IPv4/IPv6	IPv4
IPv6	IPv4/IPv6	IPv6
IPv6	IPv4	The link cannot be set up.
IPv4	IPv6	The link cannot be set up.

IP Protocol Version for the eNodeB or gNodeB	IP Protocol Version for the Peer Device (Such as the Core Network and eNodeB/gNodeB)	IP Protocol Version for Link Setup
IPv4/IPv6	IPv4/IPv6	For details about how to configure both the local and peer ends of a service interface as dual-stack nodes, see: ● LTE: *S1 and X2 Self-Management*, *eX2 Self-Management*, and *IP eRAN Engineering Guide* ● 5G NSA: *X2 and S1 Self-Management in NSA Networking* and *eXn Self-Management* ● 5G SA: *NG and Xn Self-Management*, *IP NR Engineering Guide*, and *eXn Self-Management*

Dual-stack networking may be used in NR deployment or LTE/NR IPv4-to-IPv6 transition scenarios. This section describes two typical dual-stack transmission networking scenarios.

- **Dual-stack transmission for different base stations**

  Different base stations use IPv4 or IPv6 single-stack transmission. When the base stations are connected to the same peer device (core network/OSS/clock server), the peer device must be configured with dual-stack transmission.
- Intra-base-station dual-stack transmission
  - **6.3 Intra-Base Station Dual-Stack Scenario 1: Dual-Stack Transmission for Different Interfaces**
  - **6.4 Intra-Base Station Dual-Stack Scenario 2: Dual-Stack Transmission for Different Peer Devices over the X2/Xn Interface**

    For example, the X2 interface between eNodeBs uses IPv4 transmission, and the X2 interface between the eNodeB and gNodeB uses IPv6 transmission.

    Base stations do not support IPv4-mapped IPv6 addresses, that is, IPv4 addresses cannot be mapped to IPv6 tunnels. In dual-stack networking, the bearer network must support dual-stack transmission. Different scenarios have different requirements for the peer core network devices, OSS, and base stations. Before network planning, you need to check whether the conditions for dual-stack networking are met and learn about the requirements for configuration planning. An interconnection test is required before the dual-stack networking can be used.

6.2 Dual-Stack Transmission for Different Base Stations

When the LTE/NR network is reconstructed from IPv4 to IPv6 or different networking modes are used in different areas, base stations may use different IP protocol versions for transmission. In this case, dual-stack transmission needs to be deployed for peer devices. For example:

- When base stations using different IP protocol versions connect to the same core network device over the S1 and NG interfaces, the core network device must support dual-stack transmission. IPv4 and IPv6 addresses and routes must be configured. Base stations using different IP protocol versions can access the network at the same time. Correct IP addresses can be selected for base stations using different IP protocol versions. The format of the delivered dual-stack IP address complies with 3GPP specifications (a 160-bit dual-stack IP address, of which the first 32 bits are an IPv4 address, and the last 128 bits are an IPv6 address). For other requirements, see the requirements for core network devices.
- The O&M channels of base stations use different IP protocol versions. For example, the O&M channel of the LTE/5G base station uses IPv6 transmission, and the O&M channel of the existing GSM/UMTS base station uses IPv4 transmission. If they need to connect to the same EMS, the EMS must support dual-stack transmission.

The bearer network must support dual-stack transmission, and IPv4 and IPv6 routes must be configured between the base station and peer device.

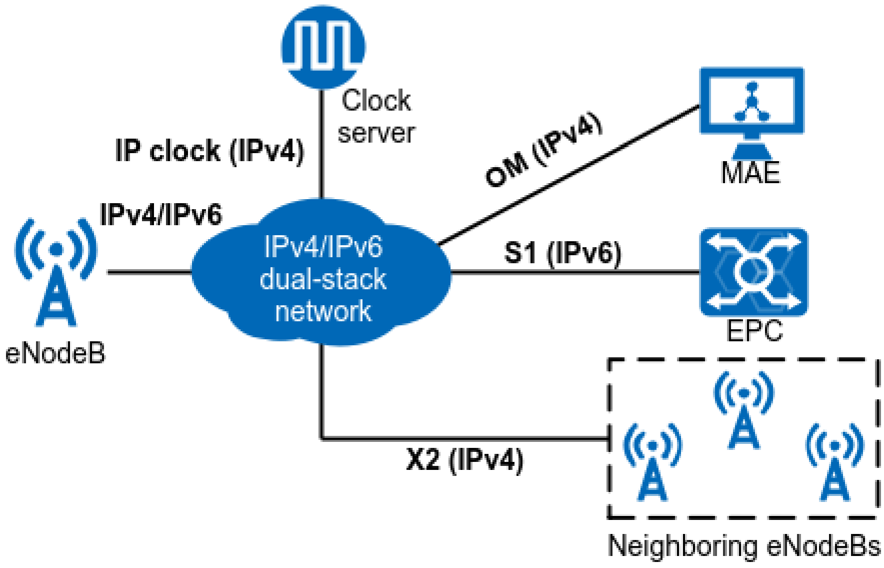
For a single base station, single-stack transmission is used. For details, see the single-stack configuration. However, if the core network device delivers dual-stack IP addresses, the base station must support the processing of the dual-stack IP addresses in 3GPP-defined format. The eNodeB/gNodeB version must be SRAN15.1 or later.

6.3 Intra-Base Station Dual-Stack Scenario 1: Dual-Stack Transmission for Different Interfaces

On an LTE/NR network, different interfaces on a base station may use different IP protocol stacks. The bearer network must use IPv4 and IPv6 routes for communication, and dual-stack transmission must be configured for the base station.

The following uses the S1 and X2 interfaces of the LTE network as an example. The S1 interface uses IPv6 transmission, and other interfaces use IPv4 transmission, as shown in **Figure 6-2**.

Figure 6-2 Example of dual-stack networking for different interfaces



Check whether the peer device meets the following conditions before network planning:

- The peer device supports IPv6 transmission or IPv4/IPv6 dual-stack transmission.
- The IPv4 and IPv6 routes between the base station and the peer device are normal.
- If the signaling from the peer device (eNodeB, gNodeB, or core network device) carries a dual-stack IP address, the IP address format must comply with 3GPP specifications (a 160-bit dual-stack IP address, of which the first 32 bits are an IPv4 address, and the last 128 bits are an IPv6 address). The eNodeB/gNodeB version must be SRAN15.1 or later. The eNodeB and gNodeB must support IPv6 transmission or can receive signaling that carries dual-stack IP addresses from the peer device.

For details about the interface self-setup requirements, see:

- LTE: *S1 and X2 Self-Management* and *eX2 Self-Management*
- NSA: *X2 and S1 Self-Management in NSA Networking* and *eXn Self-Management*
- SA: *NG and Xn Self-Management* and *eXn Self-Management*

Table 6-2 describes the dual-stack configuration requirements for the eNodeB and gNodeB.

Table 6-2 Configuration requirements for dual-stack transmission over different interfaces

Configuration Item	Base Station Configuration Requirement	Remarks
Ethernet port	• IPv4 transmission and IPv6 transmission can share physical ports or do not share physical ports. • IPv4 transmission and IPv6 transmission have the same requirements for physical layer configurations.	None
Data link layer	• When IPv4 and IPv6 share an Ethernet port, IPv4 and IPv6 can share a VLAN. If they share a VLAN, they can share a VLAN interface. Otherwise, different VLAN interfaces need to be configured. • When IPv4 and IPv6 do not share the same Ethernet port, different Ethernet ports can be connected to different Ethernet networks. The VLAN IDs of the interfaces of different ports can be the same or different. It is recommended that different VLAN IDs be configured. **NOTE** For details about the VLAN configuration requirements, see **4.3.3 VLAN**. For details about the interface configuration requirements, see **4.3.2 Interface**. • The MTUs for IPv4 and IPv6 are configured separately based on the IPv4 and IPv6 transport network plans. For details, see **4.4.4 IP Layer Fragmentation**.	None
IP layer	Interface IP addresses, service IP addresses, and static routes must be configured separately for IPv4 and IPv6 transmission. IP addresses and routes for IPv4 and IPv6 transmission are independent of each other.	None

Configuration Item	Base Station Configuration Requirement	Remarks
Transport layer	Parameters in the **SCTPHOST**, **SCTPPEER**, **USERPLANEHOST**, and **USERPLANEPEER** MOs, and the OM channel or IP clock link must be configured. IPv4 addresses are used for IPv4 transmission, and IPv6 addresses are used for IPv6 transmission.	For details, see *Transmission Resource Management* for LTE. For details, see *Transmission Resource Management* for NR.

6.4 Intra-Base Station Dual-Stack Scenario 2: Dual-Stack Transmission for Different Peer Devices over the X2/Xn Interface

During NR/LTE base station deployment or IPv4-to-IPv6 transition, some base stations use IPv6 transmission, and some use IPv4 transmission. The X2/Xn interfaces of different base stations use different IP protocol versions.

For example, in the NSA architecture, the X2 interface between the gNodeB and eNodeB uses IPv6 transmission, the X2 interface between eNodeBs uses IPv4 transmission, and the X2 interface between gNodeBs uses IPv6 transmission, as shown in **Figure 6-3**. In this case, IPv4/IPv6 dual-stack transmission needs to be configured for the X2 interface of the gNodeB.

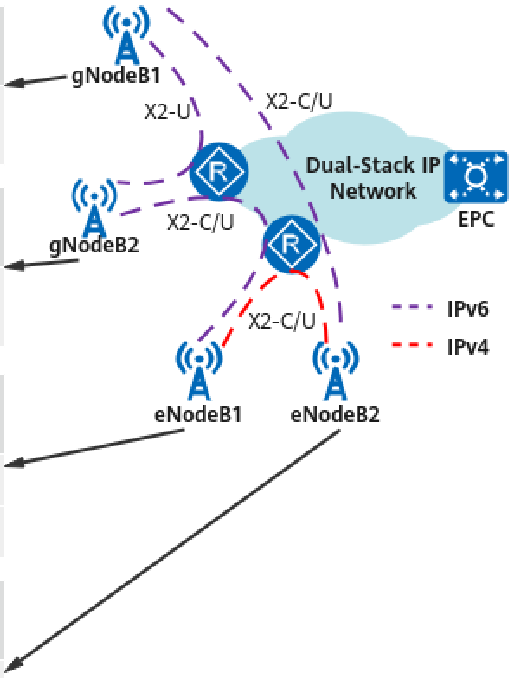
Figure 6-3 Example of IPv4/IPv6 dual-stack transmission for different peer devices over the same interface

IPVERPREFERENCE	X2-C/U local IP	Note
IPv6	2001:db8:1:1::10/64	gNB<->eNB, gNB<->gNB

IPVERPREFERENCE	X2-C/U local IP	Note
IPv6	2001:db8:1:1::15/64	gNB<->eNB, gNB<->gNB

IPVERPREFERENCE	X2-C/U local IP	Note
IPv4	10.10.11.1/24	eNB<->eNB
	2001:db8:1:2::2/64	eNB<->gNB

IPVERPREFERENCE	X2-C/U local IP	Note
IPv4	10.10.11.10/24	eNB<->eNB
	2001:db8:1:2::3/64	eNB<->gNB



The following conditions must be met in this scenario:

- The bearer network must meet the requirements described in **6.3 Intra-Base Station Dual-Stack Scenario 1: Dual-Stack Transmission for Different Interfaces**.
- The version of the neighboring eNodeB or gNodeB that is connected to the gNodeB through an X2 link must be SRAN15.1 or later.
- IPv4/IPv6 dual-stack addresses are configured for the local ends of the user plane and control plane of the X2 interface on the eNodeB.
- IPv6 single-stack addresses are configured for the local ends of the user plane and control plane of the X2 interface on the gNodeB.
- The dual-stack transmission base station meets the configuration requirements described in **Table 6-3**.

Table 6-3 Configuration requirements of IPv6 transmission between the eNodeB and the gNodeB using dual-stack transmission over the X2 interface and IPv4 transmission between eNodeBs (NSA networking)

Configuration Item	eNodeB Configuration Requirement	gNodeB Configuration Requirement	Remarks
Ethernet port	The configuration requirements are the same as those described in **6.3 Intra-Base Station Dual-Stack Scenario 1: Dual-Stack Transmission for Different Interfaces**.	IPv4 transmission and IPv6 transmission have the same requirements for physical layer configurations.	-
Data link layer	The configuration requirements are the same as those described in **6.3 Intra-Base Station Dual-Stack Scenario 1: Dual-Stack Transmission for Different Interfaces**.	For details about the VLAN configuration requirements, see **4.3.3 VLAN**. For details about the interface configuration requirements, see **4.3.2 Interface**.	-
IP layer	The configuration requirements are the same as those described in **6.3 Intra-Base Station Dual-Stack Scenario 1: Dual-Stack Transmission for Different Interfaces**.	Interface IP addresses, service IP addresses, and static routes must be configured for IPv6 transmission.	-

Configuration Item	eNodeB Configuration Requirement	gNodeB Configuration Requirement	Remarks
Transport layer	IPv4 and IPv6 user-plane and control-plane configurations for the X2 interface: ● X2-C between eNodeBs: Configure the **SCTPHOST** and **SCTPPEER** MOs for IPv4. ● X2-C between the eNodeB and gNodeB: Configure the **SCTPHOST** and **SCTPPEER** MOs for IPv6. ● X2-U between the eNodeB and gNodeB: Configure the **USERPLANEHOST** and **USERPLANEPEER** MOs for IPv6. ● X2-U between eNodeBs: Configure the **USERPLANEHOST** and **USERPLANEPEER** MOs for IPv4. ● Endpoint group: Set the **IPVERPREFERENCE** parameter in the **EPGROUP** MO for the X2 interface to **IPv4**. ● Add the control-plane and user-plane configurations of IPv4 and IPv6 addresses to the	IPv6 user-plane and control-plane configurations for the X2 interface: ● X2-C between the eNodeBs: Configure the **SCTPHOST** and **SCTPPEER** MOs for IPv6. ● X2-U between eNodeBs or between the eNodeB and gNodeB: Configure the **USERPLANEHOST** and **USERPLANEPEER** MOs for IPv6. ● Endpoint group: Set the **IPVERPREFERENCE** parameter in the **EPGROUP** MO for the X2 interface to **IPv6**. ● Add the control-plane and user-plane configurations of IPv6 addresses to the **EPGROUP** MO for the X2 interface.	● When the X2/Xn interface uses dual-stack transmission, only one IPv4 control-plane host and one IPv6 control-plane host can be configured and added to the endpoint group of the corresponding interface. Multiple user-plane hosts, user-plane peers, or control-plane peers can be configured for IPv4 and IPv6. ● For details, see *Transmission Resource Management* in *eRAN Feature Documentation* and *Transmission Resource Management* in *5G RAN Feature Documentation*.

Configuration Item	eNodeB Configuration Requirement	gNodeB Configuration Requirement	Remarks
	EPGROUP MO for the X2 interface. For details about the interface self-setup requirements in dual-stack scenarios, see *S1 and X2 Self-Management* in *eRAN Feature Documentation*.		

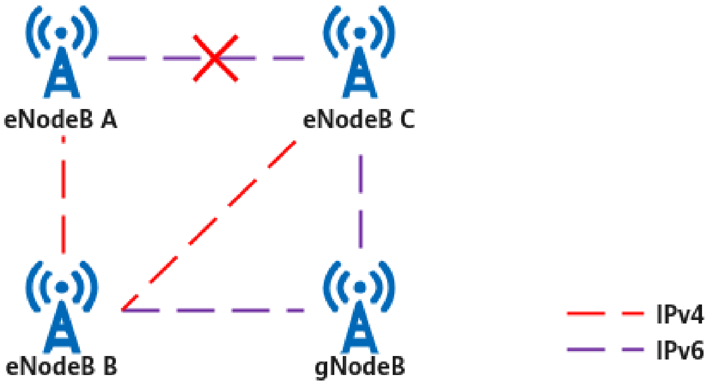
When a dual-stack IP address is configured for the X2/Xn interface, the networking requirements are as follows:

- The X2/Xn interface supports only the same IP protocol for transmission on the user plane and control plane. When you manually configure the **SCTPPEER** MO for the control plane or the **USERPLANEPEER** MO for the user plane, do not configure all the dual-stack IP addresses of the peer base station. Otherwise, the IP protocol on the control plane may be different from that on the user plane, and the X2/Xn interface cannot work properly.
- The same IP protocol is used over the X2/Xn interfaces between base stations for which all the SCTP host IP addresses or user-plane host IP addresses are dual-stack IP addresses. This is controlled by the **EPGROUP**.***IPVERPREFERENCE (LTE eNodeB, 5G gNodeB)*** parameter. For example, if the control- or user-plane host IP addresses of base stations A, B, and C are dual-stack addresses, the X2/Xn interfaces between A, B, and C must use the same IP protocol for transmission. As shown in **Figure 6-4**, the **EPGROUP**.***IPVERPREFERENCE*** parameter for the X2 interface of eNodeB C is set to **IPv6**, which is different from those of eNodeB A and eNodeB B. As a result, the X2 interface between eNodeB C and eNodeB A cannot work properly. The gNodeB uses IPv6 single-stack transmission.

Figure 6-4 Example of IPv4 transmission over some X2 interfaces between eNodeBs and IPv6 transmission over some other X2 interfaces

X2 local IP		IPVERPREFERENCE
IP1	10.10.10.12	IPv4
IP2	2001:db8::105	IPv4

X2 local IP		IPVERPREFERENCE
IP1	10.10.10.25	IPv6
IP2	2001:db8::102	IPv6



X2 local IP		IPVERPREFERENCE
IP1	10.10.10.18	IPv4
IP2	2001:db8::120	IPv4

X2 local IP	
IP2	2001:db8::100

6.5 Intra-Base Station Dual-Stack Scenario 3: Dual-Stack Transmission on the User Plane and Control Plane of the S1/NG Interface

NOTE

This scenario is complex and not recommended. It is recommended that the user plane and control plane of the core network evolve to IPv6 at the same time.

Typical scenarios of IPv6 transmission over the S1 and NG interfaces are as follows:

New NR base station deployment:

- In SA networking, it is recommended that the same IP protocol be used on the control plane and user plane of the NG interface.
- In NSA networking, the S1-U interface of the gNodeB uses IPv6 transmission, and the S1 interface of the existing eNodeB still uses IPv4 transmission. In this case, the user plane and the control plane use different IP protocols. As shown in **Figure 6-5**, in NSA networking, the S1-U interfaces of gNodeBs use IPv6 transmission, and the S1-U and S1-C interfaces of eNodeBs use IPv4 transmission.

If the LTE network evolves from IPv4 to IPv6, it is recommended that the S1-U and S1-C interfaces evolve to IPv6 at the same time.

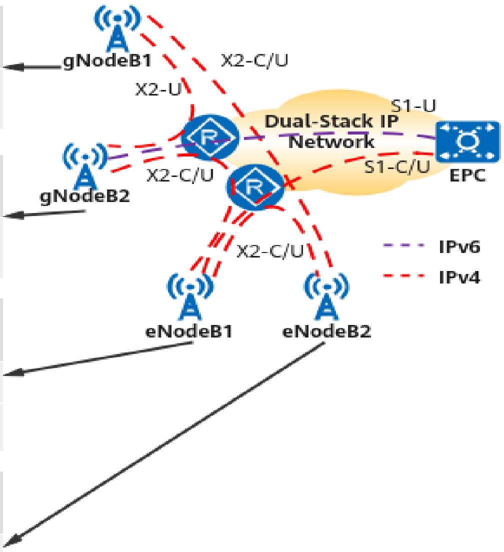
Figure 6-5 Example of IPv6 transmission over the gNodeB S1-U interfaces and IPv4 transmission over the eNodeB S1-C/S1-U interfaces (NSA networking)

IPVERPREFERENCE	S1-U local IP	Note
IPv6	2001:db8:1:1::15/64	gNB<->SGW

IPVERPREFERENCE	S1-U local IP	Note
IPv6	2001:db8:1:1::16/64	gNB<->SGW

IPVERPREFERENCE	S1-C/U local IP	Note
IPv4	10.10.11.1/24	eNB<->MME
		eNB<->SGW

IPVERPREFERENCE	S1-C/U local IP	Note
IPv4	10.10.11.10/24	eNB<->MME
		eNB<->SGW



The following conditions must be met in this scenario:

- The requirements for the bearer network are the same as those described in **6.3 Intra-Base Station Dual-Stack Scenario 1: Dual-Stack Transmission for Different Interfaces**.
- If core network devices need to support different IP protocol versions on the user plane and control plane for the same user. IPv4 and IPv6 addresses and routes for the control plane and user plane must be planned separately.
- If the signaling from the core network carries a dual-stack IP address, the IP address format must comply with 3GPP specifications (a 160-bit dual-stack IP address, of which the first 32 bits indicate an IPv4 address, and the last 128 bits are an IPv6 address). The eNodeB/gNodeB must be of SRAN15.1 or later to support IPv6 transmission or receive signaling that carries dual-stack IP addresses from the peer device.
- The S1 interface of the eNodeB uses the IPv4 single stack transmission, and the S1 interface of the gNodeB uses the IPv6 single-stack transmission.

For an IPv4-to-IPv6 transition over the S1 or NG interface for base stations, it is recommended that both the user plane and control plane use the same IP protocol version. Otherwise, interface self-setup will fail. For details, see *S1 and X2 Self-Management* in *eRAN Feature Documentation* and *NG and Xn Self-Management* in *5G RAN Feature Documentation*.

Table 6-4 Configuration requirements for dual-stack transmission on the user plane and control plane

Configuration Item	Base Station Configuration Requirement	Remarks
Ethernet port	The configuration requirements are the same as those described in **6.3 Intra-Base Station Dual-Stack Scenario 1: Dual-Stack Transmission for Different Interfaces**.	None
Data link layer	The configuration requirements are the same as those described in **6.3 Intra-Base Station Dual-Stack Scenario 1: Dual-Stack Transmission for Different Interfaces**.	None
IP layer	The configuration requirements are the same as those described in **6.3 Intra-Base Station Dual-Stack Scenario 1: Dual-Stack Transmission for Different Interfaces**.	None
Transport layer	Parameters in the **SCTPHOST**, **SCTPPEER**, **USERPLANEHOST**, and **USERPLANEPEER** MOs must be configured. IPv4 addresses are used for IPv4 transmission, and IPv6 addresses are used for IPv6 transmission.	For details, see *Transmission Resource Management* in *eRAN Feature Documentation* and *Transmission Resource Management* in *5G RAN Feature Documentation*.

7 Transmission Reliability

7.1 Introduction

Table 7-1 lists the reliability features supported by a base station in IPv6 transmission networking.

Table 7-1 Reliability features in IPv6 transmission networking

Port/Link Backup	Protocol Layer	Base Station Transmission Reliability	Support for IPv6 Transmission
Ethernet port/ link backup	Physical layer	-	-
	Data link layer	Ethernet Link Aggregation	Supported
	IP layer	IP Route Backup	Supported
Service link backup	Application layer	(Control-plane) SCTP Multihoming	Supported
		(Management-plane) OM Channel Backup	Supported

7.2 Control-Plane SCTP Multihoming

For details about SCTP, see **4.5 Transport Layer**.

If an SCTP endpoint has multiple destination addresses, the SCTP endpoint is a multi-homing endpoint.

The base station supports the following multi-homing scenarios:

- Both SCTP endpoints are multi-homed, and only parallel paths are supported. See **Figure 7-1**. The path between the first local IP address and the first peer

IP address serves as the primary SCTP path by default. The path between the second local IP address and the second peer IP address serves as the secondary SCTP path. Both primary and secondary transmission paths must be set up. If a peer address is unreachable, ALM-25955 SCTP Link IP Address Unreachable is reported.

- An endpoint of an SCTP association is multi-homed (see **Figure 7-2**). When the SCTP path between the local IP address and the first peer IP address is faulty, the SCTP path to the second peer IP address is used. This requires that the peer devices, such as the base station controller, core network, or base station, support this multi-homing mode.
  - The base station checks multiple destination IP addresses at the same time only when the peer SCTP endpoint is multi-homed. The base station reports ALM-25955 SCTP Link IP Address Unreachable only when some destination IP addresses are unreachable.
  - If the peer SCTP endpoint is a single IP endpoint, the local end checks the path from the first IP address to the peer IP address by default. If the path is unreachable, the local end checks the path from the second IP address to the peer IP address. In this scenario, the base station does not report ALM-25955 SCTP Link IP Address Unreachable.

The base station only allows multiple IP addresses of SCTP multi-homed endpoints to use the same IP protocol version. Dual-stack IP addresses cannot be configured on one endpoint.

In endpoint configuration mode, the **SCTPTEMPLATE**.***SWITCHBACKFLAG (5G gNodeB, LTE eNodeB)*** parameter specifies whether services are switched back to the primary path after it becomes functional. The value **ENABLE** is recommended.

Figure 7-1 Scenario where the two endpoints of an SCTP association are both multi-homed and only parallel paths are supported

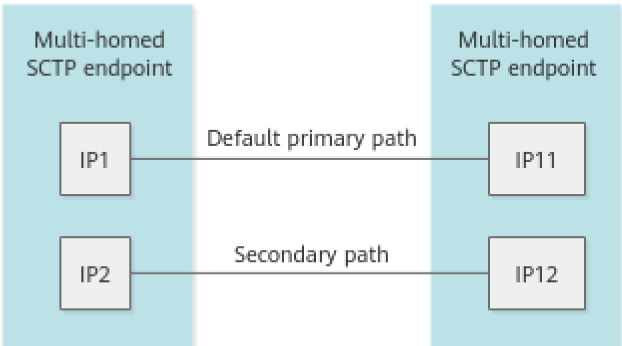
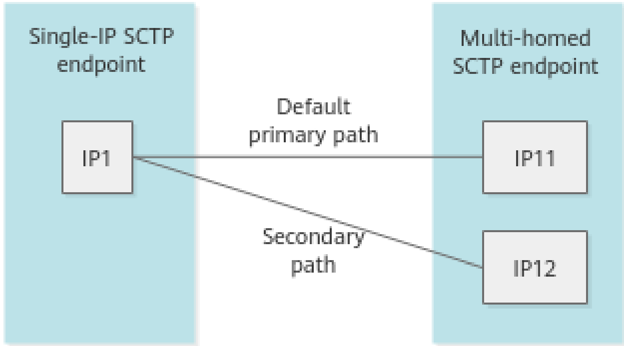


Figure 7-2 Scenario where one SCTP association endpoint is multi-homed



NOTE

If both endpoints of an SCTP association are multi-homed, configure SCTP links on the base station side as follows:

The first local IP address, second local IP address, first peer IP address, and second peer IP address are all set to valid IP addresses.

If one endpoint of an SCTP association is multi-homed, configure SCTP links on the base station side as follows:

- If the peer SCTP endpoint is multi-homed, the first local IP address is set to a valid IP address, the second local IP address is set to 0:0:0:0:0:0:0:0, and the first and second peer IP addresses are set to valid IP addresses.
- If the peer SCTP endpoint is a single IP endpoint, the first and second local IP addresses are set to valid IP addresses, the first peer IP address is set to a valid IP address, and the second peer IP address is set to 0:0:0:0:0:0:0:0.

SCTP multi-homing is not equivalent to multiple Transport Network Layer Associations (TNLAs). SCTP multi-homing is used only for reliability. Only one SCTP link is generated, and only one path to a peer IP address takes effect. In the case of multi-TNLA, multiple SCTP links are generated and all SCTP links take effect at the same time. For details about multi-TNLA, see *NG and Xn Self-Management Feature Parameter Description*.

7.3 User Plane Backup

User plane backup includes active/standby user plane and user-plane load sharing. The eNodeB, gNodeB, and LTE and NR co-MPT base station support user plane backup.

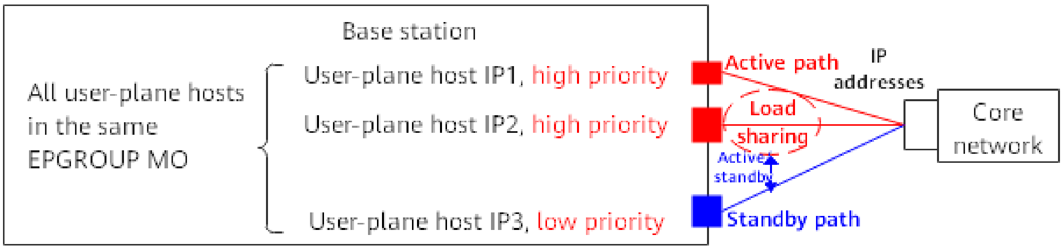
- When the transport network provides active/standby transmission path, the base station can use the active/standby user plane function to support user plane E2E redundancy backup. The backup function does not depend on the active/standby IP route of the transport network.
- When the transport network provides multiple transmission paths for load sharing, the base station can use the user-plane load sharing function to support user plane E2E load sharing, improving transmission bandwidth and enhancing reliability.

NOTE

- The active/standby user plane and user-plane load sharing can be configured only in endpoint mode.
- Currently, the S1 and NG interfaces support active/standby user plane and user-plane load sharing. The X2, eX2, Xn, and eXn interfaces support only user-plane load sharing and do not support active/standby user plane.
- Both IPv4 transmission and IPv6 transmission support active/standby user plane and user-plane load sharing, and source-based routes need to be configured.
- Currently, only the UMPT board supports user-plane load sharing.

As shown in **Figure 7-3**, for all user-plane hosts (**USERPLANEHOST** MOs) that are added to the same **EPGROUP** MO, load sharing is implemented among the user-plane hosts with the same priority, and active/standby user plane is implemented among the user-plane hosts with different priorities.

Figure 7-3 User plane backup for a base station



The base station uses the GTP-U to detect the connectivity of active and standby user-plane paths.

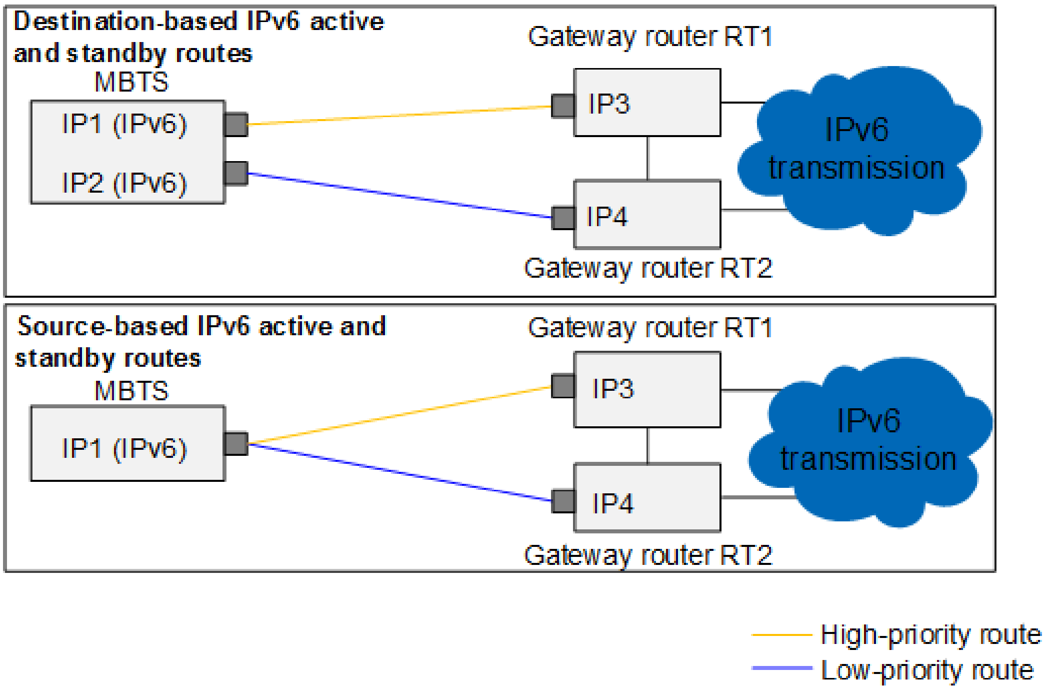
- When the base station detects that the active path is faulty but the standby path is normal, new services are established on the standby path.
- When the base station detects that the standby path is faulty, new services are established on the active path and services on the standby path are not switched to the active path.

7.4 IP Route Backup

As shown in **Figure 7-4**, the base station supports both source-based and destination-based routes in IP route backup.

- For destination-based routes, two routes with the same destination IP address but different next-hop IP addresses and priorities must be configured in active/standby mode.
- For source-based routes, two routes with the same source IP address but different next-hop IP addresses and priorities must be configured in active/standby mode.

Figure 7-4 Active and standby IP routes of the base station



Configure the logical IP routes to the base station on the gateway routers RT1 and RT2.

- On RT1, configure two routes to the base station with the next hops as the port IP1 and RT2.
- On RT2, configure two routes to the base station with the next hops as the port IP2 and RT1.

The route priority determines whether a route is active or standby. The principles for activating route priorities are as follows:

- The high-priority route is preferentially activated.
- If this route is faulty, the base station activates the other route. After being restored, the high-priority route is automatically activated.

NOTE

- IPv6 does not support equal-cost routes.
- Active and standby routes are not supported in IPv6 DR redundancy scenarios.
- Only two source-based routes with the same route type (specified by **SRCIPROUTE6**.***RTTYPE (5G gNodeB, LTE eNodeB)***) can form active and standby routes.
- Active and standby IPv6 routes (destination- or source-based routes) cannot be configured between an IPv4 IPsec tunnel and an IPv6 IPsec tunnel.

In this scenario, the base station is connected to two gateway routers, which work in active/standby mode. In the versions earlier than SRAN17.0, the active and standby IPv6 routes do not support BFD-based route switchovers. They can be switched only based on the physical port status. In SRAN17.0, the active and standby IPv6 routes bound to IPv6 BFD can be switched over.

NOTE

- If RT1 becomes faulty, although its BFD session with the base station can become normal soon after a restart, RT1 needs time to learn its routes to peer service NEs, such as core network NEs. Therefore, route switchover delay is specified on the base station so that a route switchback can be performed after RT1 has learned the routes to its peer service NEs.
- The base station does not support inter-board route backup.
- The **GTRANSPARA.*SBTIME (5G gNodeB, LTE eNodeB)*** parameter setting (specifying the time of an active/standby route switchback) is valid only when the active and standby routes are bound to a BFD session, and does not apply when the active and standby routes are switched based on the physical port status.

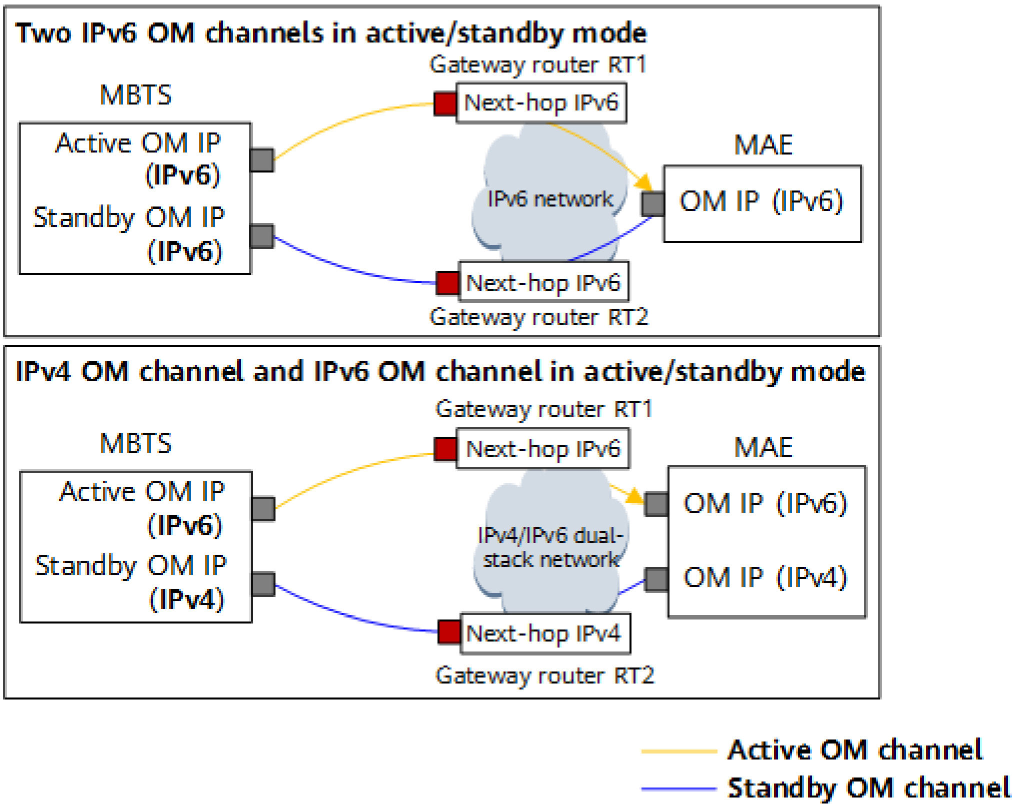
7.5 OM Channel Backup

On the base station side, OM channel backup is introduced between the base station and the MAE to provide redundancy backup for OM channels in scenarios such as hybrid transmission of high-QoS and low-QoS links and hybrid transmission of security and non-security domains. The gNodeB, eNodeB, and co-MPT base station support IPv6 OM channel backup.

As shown in **Figure 7-5**, IPv6 OM channel backup supports the following modes:

- IPv6 single-stack networking: Two OM channels for IPv6 transmission can work in active/standby mode.
- IPv4/IPv6 dual-stack networking: An OM channel for IPv4 transmission and an OM channel for IPv6 transmission can work in active/standby mode.

Figure 7-5 OM channel backup for the base station



The active and standby OM channels of the base station can share an IP address with another interface or use a separate IP address. In addition, the route from the base station to the MAE must be planned. The OM IP address and route for each base station needs to be configured on the MAE.

IP addresses used by the active and standby OM channels:

- Base station: Two IP addresses need to be configured as the local IP addresses of the active and standby OM channels.
- MAE: Only one IP address needs to be configured as the peer IP address of the active and standby OM channels.

  NE attribute parameters, IP address 1 and IP address 2, need to be configured on the MAE. The two IP addresses are the two OM IP addresses of the NE, that is, the two local IP addresses of the active and standby OM channels on the base station. The MAE automatically sets up OM channels with the two IP addresses. Therefore, when an NE is disconnected, the MAE automatically switches between IP address 1 and IP address 2 to ensure OM channel communication. OM channel communication can be ensured as long as either IP address is valid. If both IP addresses are reachable, the MAE preferentially uses IP address 1. The MAE does not automatically switch to the other IP address when the NE connection is normal.

For the IP routes used by the active and standby OM channels, in IPv6 transmission scenarios, the active and standby OM channels can be bound to the routes or not bound to the routes.

- In route-binding mode, if two OM channels for IPv6 transmission work in active/standby mode and destination-based routes are configured, you need to configure separate **IPROUTE6** MOs for the OM channels. In addition, specify **OMCH**.***BEAR (LTE eNodeB, 5G gNodeB)***, set **OMCH**.***BRT (LTE eNodeB, 5G gNodeB)*** to **YES**, and enter the indexes of the routes to be bound for **OMCH**.***RTIDX (LTE eNodeB, 5G gNodeB)***. If the MAE uses the remote HA system and different routes are required by the active and standby MAE servers, bind routes to the active and standby MAE servers through the active and standby OM channels.

  After the base station is started, the route bound to the active OM channel takes effect when the active OM channel takes effect, and the standby OM channel and the route bound to the standby OM channel do not take effect. After services are switched from the active OM channel to the standby OM channel, the route bound to the standby OM channel takes effect when the standby OM channel takes effect. Then, the active OM channel and the route bound to the active OM channel do not take effect.

  Therefore, ensure that the route bound to the OM channel forwards only the OM channel traffic, not other service flows. Otherwise, when the OMCH does not take effect, the route bound to the OMCH does not take effect, affecting forwarding of other service flows.

- In route-unbinding mode, if two OM channels for IPv6 transmission work in active/standby mode and source-based routes are configured, or if a single OM channel for IPv4 transmission and a single OM channel for IPv6 transmission work in active/standby mode, no route needs to be bound during OM channel configuration. Instead, routes for the OM channels are added by adding destination- or source-based routes, and **OMCH**.***BRT (LTE eNodeB, 5G gNodeB)*** is set to **NO**.

NOTE

- The active and standby OM channels must correspond to the same MAE-Access.
- When the active and standby OM channels are bound to routes, the active and standby OM channels cannot share the routes with services. This is because the route of the active or standby OM channel becomes unavailable if the active and standby OM channels are switched.
- For active and standby IPv6 OM channels, the MAE software version must be SRAN16.1 or later. Otherwise, OM channel backup is not supported in IPv6 transmission scenarios.
- When a base station is configured with two main control boards, the active and standby OM channels cannot be configured on different main control boards in IPv6 transmission scenarios.
- If the OM channel for IPv4 single-stack transmission and that for IPv6 single-stack transmission work in active/standby mode, IPv4 transmission supports only the new model.

The MAE detects the OM channel connectivity status by using a handshake mechanism at the application layer. If the active OM channel fails, the MAE instructs the base station through the standby OM channel to initiate an OM channel switchover.

After the active OM channel recovers, the OM channel is automatically switched back to the active OM channel, which is called the OM channel switchback function.

- When the **GTRANSPARA**.***OMCHSWITCHBACK (LTE eNodeB, 5G gNodeB)*** parameter is set to **ENABLE** to enable the OM channel switchback function, the **OMCH**.***CHECKTYPE (LTE eNodeB, 5G gNodeB)*** parameter of the active OMCH must be set to **AUTO_UDPSESSION**. When the OSS connects to the standby OMCH, the base station automatically creates a UDP session from the local IP address of the active OMCH to the MAE, and the OM channel switchback function depends on status detection of this automatically created UDP session. After detecting that the active OM channel recovers, the base station disconnects the TCP connection of the standby OM channel, which enables the MAE to initiate a connection on the active OM channel again.

  If the **PACKETFILTERING** MO is configured for the base station, ACL rules (defined by the **ACLRULE6** MO) must be manually configured for automatically established UDP sessions in the **ACL6** MO associated with the **PACKETFILTERING** MO. Otherwise, the OM channel switchback function does not take effect.

  On the base station, manually configure the **ACLRULE6** MO corresponding to the UDP session to allow the MAE packets to reach the base station. The configuration requirements are as follows:

  – Set the source IPv6 address (specified by **ACLRULE6**.***SIP (LTE eNodeB, 5G gNodeB)***) of the ACL rule to the IPv6 address of the MAE.
  – Set the destination IPv6 address (specified by **ACLRULE6**.***DIP (LTE eNodeB, 5G gNodeB)***) of the ACL rule to the local IPv6 address of the active OMCH on the base station.
  – Set the protocol type (specified by **ACLRULE6**.***PT (LTE eNodeB, 5G gNodeB)***) to **UDP**, source port 1 (specified by **ACLRULE6**.***SPT1 (LTE eNodeB, 5G gNodeB)***) to **31252**, and destination port 1 (specified by **ACLRULE6**.***DPT1 (LTE eNodeB, 5G gNodeB)***) to **65043**.

  – Set **ACLRULE6**.***ACLID (LTE eNodeB, 5G gNodeB)*** of the ACL rule to be the same as the value of **PACKETFILTERING**.***ACLID (LTE eNodeB, 5G gNodeB)***.

- When **GTRANSPARA**.***OMCHSWITCHBACK (LTE eNodeB, 5G gNodeB)*** is set to **DISABLE** to disable the OM channel switchback function, neither the MAE nor the base station automatically switches back to the active OM channel after the active OM channel recovers. The MAE initiates a connection on the active OM channel again only after the standby OM channel becomes faulty.

  A security mechanism is introduced that allows the MAE or base station to perform authentication on OMCH switchover messages transmitted by the MAE over the active or standby OM channel. This helps prevent a spoofed MAE from delivering fake switchover instructions and protect switchover messages against replay attacks or from being tampered with.

7.6 Ethernet Link Aggregation

Link aggregation (IEEE802.3ad) combines two or more Ethernet links that are added to the same link aggregation group (LAG) into a logical link. If a link in a LAG working in active/standby mode is faulty, services on this link will be switched over to functional links, preventing service interruption and improving reliability. When a LAG works in load sharing mode, the transmission link rate can be increased. However, the rate of the LAG cannot exceed the board capability specifications. Currently, Huawei base stations support only intra-board load sharing in link aggregation.

7.6.1 Basic Principles

The parameters of member links in a LAG, such as the optical/electrical mode, rate, and duplex mode, must be the same. In addition, the number of links, rate, and duplex mode at the local end must be the same as those at the peer end of the LAG. Each member port in a LAG must be configured with a priority (specified by **ETHTRKLNK.*PRI (LTE eNodeB, 5G gNodeB)***). Port priorities are used to determine the ports to be used during Link Aggregation Control Protocol (LACP) negotiation. The MAC address of a LAG is the MAC address of the port first added to the group.

The type of a LAG can be set to static or manual by setting the **ETHTRUNK.*LACP (LTE eNodeB, 5G gNodeB)*** parameter.

- Member ports are manually added to a static LAG, but LACP is used between the two ends to negotiate the actual member ports.
- Member ports are manually added to a manual LAG, and all these ports are actual member ports without LACP negotiation.

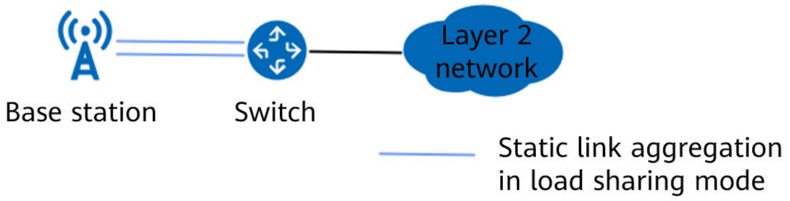
The LAG type must be consistent between the two ends.

According to the IEEE 802.3ad standard, the intervals for sending LACPDUs are classified into long timeout and short timeout. The base station supports short timeout, and its peer end must also support short timeout to enable fast link fault detection.

7.6.2 Application on the Base Station

The typical application scenario of link aggregation on the base station side is that multiple FE/GE/10GE/25GE/50GE ports on the main control board or transmission extension board of a base station form a link aggregation group and connect to a Layer 2 or Layer 3 transmission device. The link aggregation mode is static and the working mode is load sharing.

Figure 7-6 Base station connecting to a Layer 2 or Layer 3 transmission device through link aggregation



The Ethernet link aggregation load balancing factor of Layer 2 packets on the base station side is source MAC address+destination MAC address.

Table 7-2 lists the load balancing factors for Layer 3 packets.

In heavy traffic scenarios, different load balancing effects can be obtained by selecting different factors. If Layer 4 or higher-layer packet information is used as the load balancing factor, traffic is evenly distributed to member ports.

Table 7-2 Ethernet link aggregation load balancing factor of Layer 3 packets on the base station side

Packet Type	Load Balancing Factor
GTP-U	If **TRANSFUNCTIONSW**.***ETHTRKLBMODE (LTE eNodeB, 5G gNodeB)*** is set to **LB_MODE_1**, the load balancing factor is source IP address+destination IP address+Flow Label. Flow Label is filled with the last 20 bits of the TEID.
	If **TRANSFUNCTIONSW**.***ETHTRKLBMODE (LTE eNodeB, 5G gNodeB)*** is set to **LB_MODE_0**, the load balancing factor is source IP address+destination IP address+DSCP.
User-plane UDP packets	If **TRANSFUNCTIONSW**.***ETHTRKLBMODE (LTE eNodeB, 5G gNodeB)*** is set to **LB_MODE_1**, the load balancing factor is source IP address+destination IP address+Flow Label. Flow Label is filled with the hash value of the source port number and destination port number.
	If **TRANSFUNCTIONSW**.***ETHTRKLBMODE (LTE eNodeB, 5G gNodeB)*** is set to **LB_MODE_0**, the load balancing factor is source IP address+destination IP address+DSCP.

Packet Type	Load Balancing Factor
Other IP packets (such as TCP and SCTP packets)	The load balancing factor is source IP address+destination IP address+DSCP.

NOTE

- A member port in a link aggregation group of the base station cannot be configured with a non-AUTOPORT transmission resource group configured using the **IPRSCGRP** MO. If a port is manually or automatically configured with a non-AUTOPORT transmission resource group, delete the transmission resource group before adding the port to a link aggregation group.
- The **TRANSFUNCTIONSW**.***ETHTRKLBMODE (LTE eNodeB, 5G gNodeB)*** parameter takes effect only on UMPT/UMDU boards in non-IPsec scenarios.

8 Transmission Maintenance and Detection

8.1 Overview

The fault detection technologies of the IPv6 transport network include **8.2 ICMPv6 Ping**, **8.3 Trace Route**, **8.4 GTP-U Echo**, **8.5 LLDP**, IP Active Performance Measurement, and IP PM, which are applied to different protocol layers. Fault detection improves network reliability.

Table 8-1 describes the mapping between maintenance and detection mechanisms and their applicable layers.

Table 8-1 Transmission reliability and maintenance and detection methods supported by the base station

Protocol Layer	Object	Maintenance and Detection Mechanism	Detection Interval
Transport layer/ Application layer	Base station OM channel	OM handshake protocol	The detection is performed once every minute. **NOTE** By default, if the MAE-Access detects three consecutive handshake failures, ALM-25901 Remote Maintenance Link Failure is reported.
	Control-plane link	SCTP	By default, the base station detects SCTP heartbeats once every 5000 ms. The detection interval is configurable. **NOTE** After detecting an SCTP link disconnection, the base station reports ALM-25888 SCTP Link Fault.

Protocol Layer	Object	Maintenance and Detection Mechanism	Detection Interval
	S1/X2/eX2/NG/Xn/eXn user-plane path	GTP-U Echo	The base station checks the connectivity of the GTP-U tunnel once every minute. **NOTE** After detecting three (default value) consecutive GTP-U tunnel disconnections, the base station reports ALM-25954 User Plane Fault.
Network layer	IP-layer link	ICMPv6 ping	By default, the base station performs the detection once every 1000 ms.
		Trace route	None
		BFD	By default, the base station detects the route reachability once every 100 ms. The detection interval is configurable within the range of 10 ms to 1000 ms. **NOTE** By default, after detecting that a route is unreachable for three consecutive times, the base station reports ALM-25899 BFD Session Fault.
		Neighbor unreachability detection	If service data is continuously transmitted on a link, the detection time is determined by the neighbor discovery reachable time, which can be configured using the **ND**.***NDREACHABLETIME (5G gNodeB, LTE eNodeB)*** parameter. If no service data is sent or heartbeat detection is not performed for a long time on a link, the base station performs the detection after the timer for a neighbor cache entry being in the Stale state expires and the entry enters the Delay state. For details, see **4.4.3.3 Neighbor Unreachability Detection**.

Protocol Layer	Object	Maintenance and Detection Mechanism	Detection Interval
	S1/NG/X2/Xn/eXn user-plane path	IP Active Performance Measurement	For details on TWAMP, see *IP Active Performance Measurement*.
		IP PM	For details on IP PM, see *IP Performance Monitor*.
Data link layer	Ethernet link	LLDP	By default, the detection is performed once every 30s. This function is configured using the parameter specifying the interval for transmitting LLDP packets, which can be set to a value in the range of 5 to 32768, in seconds.

8.2 ICMPv6 Ping

Different from other fault detection technologies, ICMPv6 ping detection requires that the peer device should support the ICMPv6 Echo response message, and that the bearer network should not filter the ICMPv6 Echo response message. In this way, E2E link fault detection is implemented. However, peer NEs may regard ICMPv6 ping packets as attack packets and discard them. As a result, the local end considers that the transmission link is faulty when it fails to receive ICMPv6 Echo response messages.

The implementation principle of ICMPv6 ping detection is as follows: A base station sends an ICMPv6 Echo request message to the peer device. After receiving the message, the peer device sends an ICMPv6 Echo response message to the base station.

The base station supports continuous detection. By continuously sending ICMPv6 ping requests to the peer end, the base station can continuously detect the connectivity of the link to the destination IPv6 addresses. This continuous ping detection function is controlled by the ***CONTPING*** parameter in the **PING6** command. Users can press **CTRL+Q** on the keyboard or run the **STP PATHCHK** command to stop continuous ping detection.

ICMPv6 ping can be used to send packets with the destination IPv6 address being a link-local address or a global unicast address to check whether the peer end is reachable.

- If the destination IPv6 address is a link-local address, specify the destination IPv6 address and local outbound interface index.
- If the destination IPv6 address is a global unicast address, specify the source IPv6 address and destination IPv6 address.

8.3 Trace Route

The IPv6 trace route functions include route reachability detection and DSCP change detection.

NOTE

IPv6 trace route can be used only when the transport network supports response to the ICMPv6 timeout messages and destination unreachable messages and the firewall on the transport network does not filter out UDP packets used by IPv6 trace route and ICMPv6 packets.

8.3.1 Route Reachability Detection

Trace route IPv6 packets pass through all routers on the path. If a router is faulty, no ICMPv6 response packet will be returned. The transmit end then knows which hop is not reachable.

Each router decreases the Hop Limit value of an IPv6 packet by 1 when routing the packet. A router will discard a packet and notify the source node if it finds out that the value of this field in the packet is 0. IPv6 trace route sends a UDP detection packet. The initial value of Hop Limit is 1. After an ICMPv6 response packet is received, Hop Limit is increased to 2. The value of Hop Limit can be gradually increased to learn the IPv6 address and connectivity of each hop on the path. The maximum value of Hop Limit is specified by the **MAXHOPLIMIT** parameter.

8.3.2 DSCP Change Detection

The process of DSCP change detection is as follows:

- A base station sends an IPv6 trace route packet with the initial value of Hop Limit being 1. Each time the IPv6 packet passes through a router, the Hop Limit value of this packet decreases by 1. When the value of Hop Limit decreases to 0, the router discards the IPv6 packet and sends an ICMPv6 Time Exceeded Message that contains the header of the received IPv6 packet to notify the transmit end that Hop Limit expires. The base station can learn the DSCP value of the packet received by the router based on the packet returned by the router and print the DSCP value.
- The base station gradually increases the Hop Limit value and prints the DSCP values received by the router hop by hop.
- By checking for inconsistencies in all received DSCP values, users can learn whether the DSCP value has been changed, and if so, which router has changed the DSCP value.

To enable DSCP change detection, run the **TRACERT6** command with ***PATHDSCPSW*** set to **ON** on the base station side.

8.4 GTP-U Echo

GTP-U echo for the eNodeB and gNodeB is classified into static GTP-U echo and dynamic GTP-U echo.

- When enabled, static GTP-U echo is performed regardless of whether there are services. In endpoint configuration mode, the base station reports ALM-25954 User Plane Fault after detecting a fault, and reports the fault to the service layer.

  In endpoint configuration mode, static GTP-U echo has a global switch, a link-level switch, and an endpoint group switch.

  – The global switch is configured using the **GTPU**.***STATICCHK (5G gNodeB, LTE eNodeB)*** parameter.

  – The endpoint group switch is configured using the **EPGROUP**.***STATICCHK (5G gNodeB, LTE eNodeB)*** parameter.

  – The link-level switch is configured using the **USERPLANEPEER**.***STATICCHK (5G gNodeB, LTE eNodeB)*** parameter.

    When **GEPMODELPARA**.***STATICCHKMODE (5G gNodeB, LTE eNodeB)*** is set to **UPPEERSTATICCHK** and **USERPLANEPEER**.***STATICCHK (5G gNodeB, LTE eNodeB)*** is not set to **FOLLOW_GLOBAL**, the link detection setting is used. When **USERPLANEPEER**.***STATICCHK (5G gNodeB, LTE eNodeB)*** is set to **FOLLOW_GLOBAL**, the global switch setting is used. By default, the link-level switch setting is the same as the global switch setting.

    When **GEPMODELPARA**.***STATICCHKMODE (5G gNodeB, LTE eNodeB)*** is set to **EPSTATICCHK**, the endpoint group switch setting is used. By default, the endpoint group switch is turned off.

NOTE

- If the **USERPLANEPEER** MO has been configured, the system automatically updates the configuration of this MO when static GTP-U echo is enabled, without changing the value of the **USERPLANEPEER**.***STATICCHK (5G gNodeB, LTE eNodeB)*** parameter.
- ALM-25954 User Plane Fault of the major severity is reported when there is at least one user-plane path fault for the same service type. ALM-25954 User Plane Fault of the critical severity is reported when all user-plane paths corresponding to the S1/NG interface are faulty.
- In the simplified endpoint mode, as the **EPGROUP** MO cannot be manually configured, static GTP-U echo does not support the endpoint group configuration mode.

- Dynamic GTP-U echo takes effect only when there are services and static GTP-U echo is disabled. When a fault occurs, dynamic GTP-U echo does not report ALM-25954 User Plane Fault (in endpoint configuration mode) but only reports the fault to the service layer.

It is recommended that static GTP-U echo be enabled. This prevents UEs from being released without any prompt information when a transmission link becomes faulty. Users can specify the timeout duration of echo frames and the number of echo frame timeouts by setting the **GTPU**.***TIMEOUTTH (5G gNodeB, LTE eNodeB)*** and **GTPU**.***TIMEOUTCNT (5G gNodeB, LTE eNodeB)*** parameters, respectively. The two parameters are globally configured and cannot be set to different values for individual links.

In endpoint configuration mode, GTP-U echo uses the local IP address of the **USERPLANEHOST** MO and the peer IP address of the **USERPLANEPEER** MO in the same **EPGROUP** MO as the source and destination IP addresses, respectively. In this case, the DSCP values of the GTP-U control packets are uniformly configured for the entire network.

NOTE

In base station IPv6 transmission, only the endpoint configuration mode is supported.

When static GTP-U echo is enabled, the mechanism of reporting X2/eX2/gNBCUX2/gNBCUXn/gNBDUeXn user-plane fault alarms is optimized to improve the accuracy of user-plane fault alarms. If a user-plane path fault is detected and there are X2/eX2/gNBCUX2/gNBCUXn/gNBDUeXn services triggering a bearer setup attempt, an X2/eX2/gNBCUX2/gNBCUXn/gNBDUeXn service alarm will be reported, enabling precise reporting of user-plane fault alarms. Precise reporting of X2/eX2/gNBCUX2/gNBCUXn/gNBDUeXn user-plane fault alarms is enabled when the **INTER_UP_ALM_PRECISE_SW** option of the **TRANSALGEXTPARA**.***TNLBEARERMNGALGSW*** parameter is selected.

NOTE

Precise reporting of eX2 user-plane fault alarms applies only to eX2 interfaces in non-ideal backhaul mode (with **EX2**.***BackhaulType*** set to **NON_IDEAL**).

8.5 LLDP

Link Layer Discovery Protocol (LLDP) is a Layer 2 discovery protocol defined in the IEEE 802.1ab standard. It allows network devices to advertise local information in the local subnet, including the system name, system description, port ID, and MAC address. An OSS can use LLDP to quickly obtain the Layer 2 network topology information and topology changes of the base station and to further obtain the topology relationship between the base station and the peer device. With this information, transmission faults can be quickly located.

The base station supports manual and automatic LLDP configurations. The LLDP configuration mode can be specified by setting **LLDPGLOBAL**.***PORTCFGMODE (LTE eNodeB, 5G gNodeB)*** to **MANUAL** or **AUTO**.

- Manual LLDP configuration

  Users can run the **ADD LLDP** command to add the local end information of an LLDP port.

  – If the peer device is configured with a VLAN, the **LLDP**.***BNDVLAN (LTE eNodeB, 5G gNodeB)*** parameter that specifies the VLAN bound to the LLDP local port must be set to **YES**. The **LLDP**.***VLANID (LTE eNodeB, 5G gNodeB)*** parameter that specifies the VLAN ID and the **LLDP**.***VLANPRI (LTE eNodeB, 5G gNodeB)*** parameter that specifies the VLAN priority must be set based on the configuration on the peer device.

  – If the peer device is not configured with a VLAN, **LLDP**.***BNDVLAN (LTE eNodeB, 5G gNodeB)*** must be set to **NO**.

- Automatic LLDP configuration

  The base station automatically configures LLDP port information based on the configured Ethernet port. The automatically configured LLDP port is not bound to any VLAN.

The **SET LLDPGLOBALINFO** command is used to set the LLDP global configuration information, including the interval for transmitting LLDP packets (**LLDPGLOBAL**.***TXINTVAL (LTE eNodeB, 5G gNodeB)***), LLDP TTL multiplier (**LLDPGLOBAL**.***HOLDMULTI (LTE eNodeB, 5G gNodeB)***), reinitialization delay

(**LLDPGLOBAL**.***REINITDELAY (LTE eNodeB, 5G gNodeB)***), LLDP packet transmission delay (**LLDPGLOBAL**.***DELAY (LTE eNodeB, 5G gNodeB)***), and base station and microwave collaboration switch (**LLDPGLOBAL**.***BTSMWCOSW (LTE eNodeB, 5G gNodeB)***). The **BKP_PSAV_EVALUATE_DATA_SW** option of the **LLDPGLOBAL**.***BTSMWCOSW (LTE eNodeB, 5G gNodeB)*** parameter specifies whether the base station transmits its backup power saving data to the microwave device. If this option is selected and the microwave device's backup power subscription switch is turned on, the base station sends the obtained backup power saving data to the microwave device through LLDP packets.

8.6 BFD

BFD is a type of high-speed independent Hello protocol and supports link fault detection in the millisecond range.

After a fault is detected, fault isolation can be triggered by association with upper-layer protocols, reducing service loss and improving system reliability.

BFD is classified into single-hop BFD (SBFD) and multi-hop BFD (MBFD).

SBFD applies to end-to-end direct connections in Layer 2 networking.

MBFD applies to end-to-end indirect connections in Layer 3 networking. Multiple intermediate nodes can exist between the two ends of an MBFD session.

8.6.1 Detection Mechanism

BFD requires the setup of a BFD session with which the BFD detects the connection between two ends.

A BFD end plays either an active role or a passive role during session initialization.

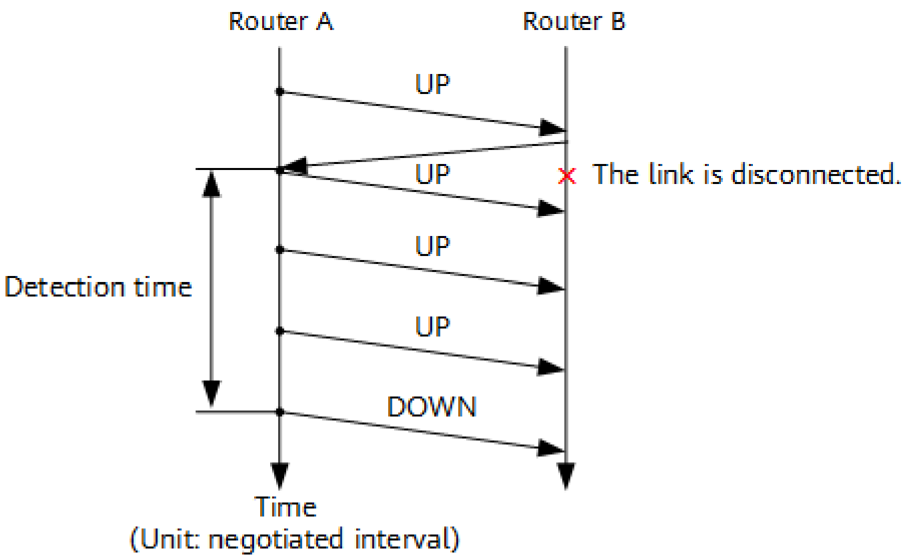
- A BFD end playing the active role can send BFD control packets for a particular session, regardless of whether it has received any BFD packets for that session.
- A BFD end playing the passive role can start sending BFD control packets for a particular session only after it receives a BFD packet for that session.

A BFD session can be established only when at least one of the devices at both ends plays the active role. The base station always plays the active role in BFD.

BFD uses BFD messages as heartbeat messages. If a BFD end transmits several BFD messages over a link but does not receive any response messages, it considers the link faulty.

Figure 8-1 illustrates BFD messages exchanged when a link is disconnected.

Figure 8-1 BFD message exchange when the link is disconnected



In asynchronous BFD mode, both BFD ends periodically send BFD control packets to each other. If the BFD control packets are received within the detection time shown in **Figure 8-1**, the session is considered **UP**. If BFD control packets are not received within the detection time, the BFD session is considered **DOWN**.

The interval for transmitting BFD control packets is proportional to Max(Minimum transmit interval, Minimum receive interval). The proportion is determined by the detection multiplier. The detection time can be calculated at the BFD transmit end by using the following formula: Detection time = Detection multiplier of the transmit end x Max(Minimum transmit interval, Minimum receive interval)

During BFD session negotiation, packets from a base station carry two fields: my discriminator and your discriminator. My discriminator is generated by the base station, and your discriminator is generated by the router. After parsing my discriminator in the packet from the router, the base station sends it as your discriminator in its own packet to the router. My discriminator of the base station is specified by the **BFDSESSION**.***MYDSCRMGENMODE*** (old model)/ **BFD**.***MYDSCRMGENMODE*** (new model) parameter.

- If the parameter is set to **STATIC_CONFIG**, my discriminator is manually set to a fixed value.
- If the parameter is set to **DYNAMIC_GEN** or **DYNAMIC_GEN_ENHANCE**, my discriminator is randomly generated. In this case, ensure that the router does not limit the discriminators on the base station side.

8.6.2 Configuration Requirements

The requirements for configuring the BFD function are as follows:

- Both ends of a BFD session must support BFD.
- BFD must be started on both ends of a BFD session, and the detection time must be equal or nearly equal.

- SBFD requires a physical port on the end that sends BFD control packets to be bound to the source and destination IP addresses of a connection to which the SBFD applies.
- MBFD requires the source IP address to be bound to the destination IP address of a connection at the end that sends BFD control packets.

  Either device IP addresses or interface IP addresses of an interface board can be used as the local IP address of an MBFD session. The peer IP address cannot be on the same segment as any of the local IP addresses.

  The link-local address cannot be used as the source or destination IP address of a BFD session.
- Detection results can be linked with upper-layer protocols.

  If BFD is linked with upper-layer protocols and a fault is detected, the active and standby ports can be switched and the route with the detection IP address of the BFD session as the next- hop IP address will be removed.
- BFD in asynchronous mode is supported and BFD in demand mode is not supported.
- If static BFD is enabled on the router and my and your discriminators are specified, the base station's generation mode of my discriminator must be static configuration, and my discriminator on the base station side must be the same as your discriminator configured on the router to avoid negotiation failures.

If BFD control packets are transmitted in plaintext, security risks such as bogus packets and replay attacks exist. Therefore, BFD authentication is introduced to support SBFD on base stations and base station controllers.

NOTE

BFD authentication complies with the RFC5880 protocol. The protocol version for BFD sessions must be standard and cannot be Draft 4.

- To enable BFD authentication for a base station, ensure that the peer router also supports BFD authentication. If the peer router does not, the authentication will fail.
- After BFD authentication is enabled, the number of configured keys and key values must be consistent between the local and peer ends. Otherwise, the authentication will fail.

Parameter configuration on the base station side:

- **BFD**.***HT (5G gNodeB, LTE eNodeB)***: used to configure the BFD hop type.
- **BFD**.***DM (5G gNodeB, LTE eNodeB)***: used to configure the detection multiplier of a BFD session.
- **BFD**.***MINTI (5G gNodeB, LTE eNodeB)***: used to configure the effective minimum transmit interval.
- **BFD**.***MINRI (5G gNodeB, LTE eNodeB)***: used to configure the effective minimum receive interval.
- **BFD**.***CATLOG (5G gNodeB, LTE eNodeB)***: used to configure whether to allow port switchovers.
- **BFD**.***MYDSCRMGENMODE (LTE eNodeB, 5G gNodeB)***: used to configure the generation mode of my discriminator.

- **BFD**.***MYDISCREAMINATOR (LTE eNodeB, 5G gNodeB)***: used to configure my discriminator.

8.6.3 Binding Relationship Between SBFD and IP Route

In the current version, only SBFD sessions can be bound to IP routes. To implement SBFD, an SBFD session must be bound to an **IPROUTE6/SRCIPROUTE6** MO.

After BFD authentication is enabled on a base station, associated route switchover is performed after BFD. Once BFD authentication fails or a link fault occurs, link switchover is triggered. The switchover procedure is the same as that after the existing BFD.

The parameters used to configure a binding relationship between an SBFD session and an IP route on the base station side are as follows:

- On the base station side, the **BFD**.***CATLOG (5G gNodeB, LTE eNodeB)*** (in the new model) parameter is used:
  - When this parameter is set to **RELIABILITY**:

    If an SBFD session detects a fault, the base station will automatically isolate the route with the peer IP address as the next-hop address.

    If active and standby routes have been configured, a route switchover will be performed after the fault is detected. After the faulty route is restored, a route switchback will take place when the time specified by **GTRANSPARA.*SBTIME (5G gNodeB, LTE eNodeB)*** elapses.
  - When this parameter is set to **MAINTENANCE**:

    If an SBFD session detects a fault, the base station will not automatically isolate the route with the peer IP address as the next-hop address.

    A route switchover will not occur even if active and standby routes have been configured.

9 Engineering Guidelines

- For details about eRAN IPv6 transmission, see *IP eRAN Engineering Guide* in *eRAN Feature Documentation*.
- For details about NR IPv6 transmission, see *IP NR Engineering Guide* in *NR Feature Documentation*.

10 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, LOFD-001016 or TDLOFD-001016.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter.

View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

11 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, LOFD-001016 or TDLOFD-001016.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

12 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

13 Reference Documents

- 3GPP TS 29.281, "General Packet Radio System (GPRS) Tunnelling Protocol User Plane (GTPv1-U)"
- *IP Performance Monitor*
- *Ethernet OAM*
- *IP Active Performance Measurement*
- *Common Transmission*
- *SRAN Networking and Evolution Overview*
- *IPv4 Transmission*
- *IPsec*
- *X2 and S1 Self-Management in NSA Networking*
- Feature parameter description documents in *eRAN Feature Documentation*:
  - *Transmission Resource Management*
  - *Synchronization*
  - *IP eRAN Engineering Guide*
  - *eX2 Self-Management*
  - *S1 and X2 Self-Management*
- Feature parameter description documents in *5G RAN Feature Documentation*:
  - *Transmission Resource Management*
  - *Synchronization*
  - *eXn Self-Management*
  - *NG and Xn Self-Management*
  - *IP NR Engineering Guide*